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A STATISTICAL FRAMEWORK OF WATERMARKS FOR LARGE LANGUAGE MODELS: PIVOT, DETECTION EFFICIENCY AND OPTIMAL RULES

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Since ChatGPT was introduced in November 2022, embedding (nearly) unnoticeable statistical signals into text generated by large language models (LLMs), also known as watermarking, has been used as a principled approach to provable detection of LLM-generated text from its human-written counterpart. In this paper, we introduce a general and flexible framework for reasoning about the statistical efficiency of watermarks and designing powerful detection rules. Inspired by the hypothesis testing formulation of watermark detection, our framework starts by selecting a pivotal statistic of the text and a secret key—provided by the LLM to the verifier—to control the false positive rate (the error of mistakenly detecting human-written text as LLM-generated). Next, this framework allows one to evaluate the power of watermark detection rules by obtaining a closed-form expression of the asymptotic false negative rate (the error of incorrectly classifying LLM-generated text as human-written). Our framework further reduces the problem of determining the optimal detection rule to solving a minimax optimization program. We apply this framework to two representative watermarks—one of which has been internally implemented at OpenAI—and obtain several findings that can be instrumental in guiding the practice of implementing watermarks. In particular, we derive optimal detection rules for these watermarks under our framework. These theoretically derived detection rules are demonstrated to be competitive and sometimes enjoy a higher power than existing detection approaches through numerical experiments.

1. Introduction. Large language models (LLMs) have emerged in recent years as a disruptive technology to generate human-like text and other media [2, 44, 60]. While this reality enhances productivity in many sectors, the mismatch between ownership and generation of content could lead to several unwanted outcomes.

Exacerbating misinformation. The ability of LLMs to generate a large amount of text in parallel can be easily leveraged to exacerbate the spread of misinformation [67, 72], which can be enabled by deploying automated bots on social media platforms [56]. It may also facilitate fraud and deception by pretending to be humans interacting with their relatives and acquaintances.

Facade of AI-assisted education. LLMs impose challenges to education because students may use powerful LLMs to write essays for themselves [42, 58]. This deprives students of opportunities to practice their own skills and creates inequalities among students depending on the capabilities of the LLMs they access.

Data pollution. The internet will soon consist of more LLM-generated text than human-written text. If LLM-generated text is indiscriminately mixed with human-written text for training, it becomes difficult to create high-quality data for developing next-generation LLMs [16, 48, 54].

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An initial effort to mitigate these problems has been to leverage specific patterns of LLM-generated text to distinguish it from human-generated text [23, 43, 73]. However, this approach has become increasingly ineffective as models such as ChatGPT-4, Claude 3, Gemini 1.5 Pro and many others have reached a level that makes it significantly difficult, if not impossible, to distinguish their generated text from human-written text [66].

A more viable approach is to watermark text by embedding a signal into the LLM-generated text in a manner that allows the watermark to be *provably* detected [1, 14]. The property of provable detection is crucial because it allows the verifier to identify LLM-generated text for malicious purposes without relying on assumptions about the text, which may not always hold. Moreover, a reasonable watermarking scheme should be approximately unbiased, meaning it does not significantly distort the meaning or style of the original text.

The necessity of watermarking LLM-generated text was highlighted in the Biden administration's October 2023 executive order, which incorporated proposals for watermarking LLM-generated text and other AI-generated content [9]. As part of the executive order, the U.S. Department of Commerce will help develop standards to watermark LLM-generated content. Accordingly, OpenAI, Google, Meta and other tech giants have pledged to develop watermarking systems [8].

This reality has made it imperative for researchers to develop watermarking methods for LLM-generated text. Within a year since 2023, numerous watermarks have been proposed [20, 22, 27, 32, 34, 40, 68, 70, 76, 77]. A common feature of these methods is leveraging the probabilistic nature of LLMs. In essence, these methods incorporate pseudorandomness into the text-generation process of LLMs, and the coupling between the LLM-generated text and the pseudorandomness serves as a signal that can be used for detecting the watermark. The constructed signal becomes pronounced for detection only when the pseudorandom numbers are known, making it practically difficult for one to remove the watermark without access to the pseudorandom numbers [14].

To understand how watermarks work for LLMs in more detail, we must first introduce the concept of "tokenization" in LLM text processing. Tokenization involves breaking down the text into smaller units called tokens, which are informally known as subwords. These tokens can be words, parts of words or even punctuation marks. For example, the sentence "Hello, world!" when tokenized, might be split into four tokens: ["Hello," ",", "world," "!"].[1] An LLM generates each token sequentially by sampling from a probability distribution conditioned on prior tokens, among other things. Typically, the size of the token vocabulary is of the order of 10^4 and varies with language models. Letting $\mathcal{W}$ denote the vocabulary of all tokens, for example, $|\mathcal{W}| = 50{,}272$ for the OPT-1.3B model [75], $|\mathcal{W}| = 32{,}000$ for the LLaMA series models [60], and $|\mathcal{W}| = 50{,}257$ for GPT-2 and GPT-3.5 series models [10, 49].

After generating text in the form of a token sequence, $w_1 w_2 \cdots w_{t-1}$, the (unwatermarked) LLM generates the next token w_t according to a multinomial distribution $\boldsymbol{P}_t := (P_{t,w})_{w\in\mathcal{W}}$ on the vocabulary $\mathcal{W}$, satisfying $\sum_{w\in\mathcal{W}} P_{t,w} = 1$.[2] We call $\boldsymbol{P}_t$ the *next-token prediction* (NTP) distribution at step t, and it depends on all prior generated tokens, the user-supplied prompt, as well as system prompts that are hidden from users [10, 49, 63].

In contrast, a watermarked LLM generates the next token that is jointly determined by a pseudorandom variable and the NTP distribution. Let ζ_t denote the pseudorandom variable at step t, which is available only to the verifier. Formally, the watermarked LLM samples a token according to the rule [27, 35, 68]

$$w_t := \mathcal{S}(\boldsymbol{P}_t, \zeta_t),$$

[1] Refer to the website https://platform.openai.com/tokenizer for user-customized examples of tokenization.

[2] As a convention, throughout this paper three subscripts appear: letter$_t$, letter$_w$ and letter$_{t,w}$, which correspond to step t, token w and token w at step t, respectively.

where $\mathcal{S}$ is a (deterministic) decoder function. To achieve approximate unbiasedness, we require that the probability distribution of $w_t \equiv \mathcal{S}(\boldsymbol{P}_t, \zeta_t)$ over the randomness[3] embodied in ζ_t is close to $\boldsymbol{P}_t$, conditional on all previous tokens, $w_1 \cdots w_{t-1}$ [27, 34]. In this regard, an unbiased watermark roughly corresponds to a sampling method from multinomial distributions.

With the generated text now indistinguishable from that of the unwatermarked LLM, it seems at first glance hopeless to obtain provable detectability for watermarked text [52, 74]. Interestingly, detection can be made possible by carefully designing the decoder $\mathcal{S}$ to impose a coupling relationship between the token and pseudorandom variable, even without knowing the NTP predictions [14].[4] In the following, we elaborate on this point by considering perhaps the simplest example of a watermark that achieves both unbiasedness and provable detectability.

A baby watermark. Envision an LLM that involves only two tokens, 0 and 1—that is, $\mathcal{W} = \{0, 1\}$. Let $\boldsymbol{P}_t = (P_{t,0}, P_{t,1})$ denote the NTP distribution at step t, and let ζ_t be i.i.d. copies of the standard uniform random variable $U(0, 1)$. Set the decoder as follows:

$$\mathcal{S}(\boldsymbol{P}, \zeta) = \begin{cases} 0 & \text{if } \zeta \le P_0, \\ 1 & \text{otherwise.} \end{cases} \tag{1}$$

This watermark is unbiased. Intuitively, if ζ_t is large, then w_t is more likely to be 1 instead of 0 and vice versa. This intuition suggests using the following statistic for detecting the watermark:

$$\sum_{t=1}^{n} (2w_t - 1)(2\zeta_t - 1), \tag{2}$$

which measures the correlation between the tokens and pseudorandom variables. When the watermark is present, this statistic tends to be larger in distribution than when the watermark is absent. One can conclude that a watermark is detected if this statistic is above a certain threshold. Despite being intuitive, however, this statistic is ad hoc, and one cannot rule out the possibility of a better detection rule.

To better understand practical watermarking schemes, we turn to real-world LLMs where the size of the token vocabulary is very large. As a recap, at the core of an unbiased watermark is a sampling method for multinomial distributions. Perhaps the two most common sampling methods are the Gumbel-max trick [29, 41] and the inverse transform [18]. Interestingly, these two sampling methods correspond precisely to two important and representative watermarks [1, 34].

Gumbel-max watermark [1]. Let $\zeta = (U_w)_{w \in \mathcal{W}}$ consist of $|\mathcal{W}|$ i.i.d. copies of $U(0, 1)$. A version of the Gumbel-max trick [24] states that $\arg\max_{w \in \mathcal{W}} \frac{\log U_w}{P_w}$ follows the NTP distribution $\boldsymbol{P} \equiv (P_w)_{w \in \mathcal{W}}$.[5] Recognizing this fact, Scott Aaronson proposed the following decoder [1]:

$$\mathcal{S}^{\text{gum}}(\zeta, \boldsymbol{P}) := \arg\max_{w \in \mathcal{W}} \frac{\log U_w}{P_w}, \tag{3}$$

[3] Strictly speaking, ζ_t has no randomness. However, modern cryptographic theories ensure that ζ_t behaves very much like a random variable. See more details in Section 2.

[4] We cannot use the probabilities $\boldsymbol{P}_t$'s as the verifier in general does not have access to the LLM that generates the tokens. Moreover, the prompt for generating the text is unavailable to the verifier, hence the verifier cannot obtain $\boldsymbol{P}_t$ even having access to the LLM.

[5] It is noteworthy that the Gumbel-max trick has a broad range of applications, including partition function estimation [25], ranking under the random utility model [55] and computational sampling in machine learning [29, 41, 46].

which is, by definition, unbiased [20, 47, 77]. This watermark has recently been implemented internally at OpenAI [1]. Aaronson suggested declaring the presence of the watermark if the following statistic is above a certain threshold: $T_n^{\mathrm{ars}} = -\sum_{t=1}^{n} \log(1 - U_{t,w_t})$. The intuition is that, when the watermark is employed, (3) implies that a token w_t is more likely to be selected when its associated pseudorandom number ζ_{t,w_t} is large. In contrast, when the text is written by a human, ζ_{t,w_t} would not be larger than other entries in the distribution at step t. Despite being intuitive, it is worth mentioning that there are countless detection rules capable of capturing this distribution shift. In particular, it is not clear whether this detection rule is optimal in any sense.

Inverse transform watermark [34]. It is well known that any univariate distribution can be sampled by applying the inverse cumulative distribution function (CDF) to $U(0,1)$. Given an NTP distribution $\boldsymbol{P}$ and π that maps all tokens in $\mathcal{W}$ to a permutation of $1, 2, \ldots, |\mathcal{W}|$, consider the multinomial distribution with probability mass $P_{\pi^{-1}(i)}$ at i for $1 \le i \le |\mathcal{W}|$. The CDF of this distribution takes the form $F(x;\pi) = \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \le x\}}$. Taking as input $U \sim U(0,1)$, the generalized inverse of this CDF is defined as $F^{-1}(U;\pi) = \min\{i : \sum_{w' \in \mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w') \le i\}} \ge U\}$, which by construction follows the multinomial distribution $\boldsymbol{P}$ after applying the permutation π. Making use of this fact, [34] proposed the inverse transform watermark with the following decoder:

$$\mathcal{S}^{\mathrm{inv}}(\boldsymbol{P}, \zeta) := \pi^{-1}\big(F^{-1}(U;\pi)\big),$$

where the pseudorandom variable $\zeta := (\pi, U)$ and the permutation π is uniformly at random. By definition, this watermark is unbiased. To detect the watermark, [34] proposed detection rules that, roughly speaking, examine the absolute difference between U_t and the rank of the generated token, $\pi_t(w_t)$, and sum the differences across the token sequence (see details in Section 4). When the text is watermarked, the difference turns out to be small because the involved two random variables are highly dependent. Similar to the Gumbel-max watermark, the detection of this watermark could potentially benefit from a more principled derivation to enhance its power.

1.1. *This paper.* From a statistical viewpoint, a watermark scheme's performance at the detection phase is evaluated by two competing criteria. The first is how likely it would mistakenly detect human-written text as LLM-generated text, and the second is the probability that it would mistakenly classify LLM-generated text as its human-written counterpart. This perspective relates watermarks to hypothesis testing, which formally calls the two aforementioned errors, Type I error and Type II error, respectively. In this regard, the effectiveness of the Gumbel-max watermark and inverse transform watermark, even including the baby watermark, is not clear yet, though [20, 47] compared their efficiency through empirical experiments. For example, even though they come with detection rules that seem intuitive, it is not clear if they are statistically optimal in the sense of having the optimal trade-off between Type I and Type II errors. If they are not optimal, it is of interest to find a better detection rule.

In general, we wish to have a general and flexible framework that can guide the development of watermarks through optimizing its detection phase and assessing watermarks in a principled manner. Specifically, the challenge lies in how we can provably control the Type I error for any given watermark, considering that the NTP distributions, which are not accessible to the verifier, vary from token to token. Once Type I error control is achieved, the next question is how to evaluate the Type II error, preferably through a closed-form expression, which also hinges on the unknown NTP distributions. Having known both the Type I and Type II errors, the final step involves comparing watermark detection rules to ultimately identify the optimal detection rule based on our knowledge of the LLM.

In this paper, we address these questions and challenges in a unified way by making the following contributions.

A Statistical framework for watermarks. A major contribution of this paper is a general statistical framework for developing statistically sound watermark detection rules through precisely evaluating the Type I and Type II errors from the hypothesis testing viewpoint. Under this framework, Type I error is controlled by leveraging a pivotal statistic that is distributionally independent of the NTP distributions under the null. This framework is accompanied by a technique for evaluating the asymptotic Type II error using large deviation theory, and moreover, relies on the notion of class-dependent efficiency to tackle the challenge of unknown and varying NTP distributions. Finally, this framework formulates the problem of finding the most powerful detection rule as a minimax optimization program.

This framework is formally developed in Section 2.

Application to the Gumbel-max watermark. We apply our framework to Aaronson's Gumbel-max watermark in Section 3. Our main finding is that Aaronson's detection rule is suboptimal in the sense that its class-dependent efficiency is relatively low. Moreover, by maximizing the class-dependent efficiency, we obtain the optimal detection rule, which admits a simple analytical expression. This optimal detection rule is shown to outperform existing rules in numerical experiments. Underlying these results is a technique that can reduce the optimality problem to a convex geometry problem for the Gumbel-max watermark, which is a contribution of independent interest to future research on watermarks.

Application to the inverse transform watermark. Next, we apply our framework to the inverse transform watermark in Section 4. Our main finding is twofold. First, we overcome a significant challenge in applying our framework to analyze this watermark by deriving an asymptotic distribution when the text is watermarked. Second, we obtain the optimal detection rule for the inverse transform watermark in a closed-form expression by maximizing its class-dependent efficiency. Our numerical experiments corroborate its efficiency.

1.2. *Related work.* The most closely related work to ours is the Gumbel-max watermark [1] and the inverse transform watermark [34]. In [20, 77], the authors introduced unbiased watermarks that are robust to probability perturbations or multi-bit processing. Other unbiased watermarks in this fast-growing line of research include [14, 27, 68]. In addition, a popular example is the red-green list watermark, which splits the vocabulary into red-green lists based on hash values of previous n-grams and slightly increases the probability of green tokens embedding the watermark [11, 31, 32, 40, 76]. In the detection phase, a high frequency of green tokens suggests the text is LLM-generated. This type of watermark is biased since the NTP distributions have been altered, thereby leading to performance degradation of the LLM.

In contrast, there is much less work on the theoretical front. A notable exception is the work of [28], which approached watermark detection from the perspective of composite independence testing. Their framework requires model providers to offer random rejection regions. In contrast, our approach employs a pivotal statistic to detect distributional shifts between human-written and LLM-generated text, allowing verifiers to choose any rejection region. While both approaches aim to minimize a form of worst-case Type II error, the optimal detection rule proposed by [28] is hindered by a nonvanishing Type II error due to the consideration of many worst-case scenarios, and it is computationally inefficient because of the exponentially large rejection regions required.[6] Conversely, our optimal rule generally achieves a vanishing Type II error by addressing fewer worst-case scenarios, benefiting from

[6]See Theorem 3.10 of [28] and the subsequent discussion.

the well-regularized null behavior of pivotal statistics, and is computationally efficient due to its sum-based structure.

Research on watermarking text has been conducted well before the advent of modern language models. This body of research focuses on watermarking text by modifying it to introduce specific patterns that are unlikely to be noticeable to readers. This includes synonym substitution [59], syntactic restructuring [4] and linguistic steganography [15]. A common weakness of these approaches lies in their biasedness and vulnerability to attacks that aim to remove watermarks [12, 78].

In a different direction, many methods have been proposed to detect text generated by LLMs—often not watermarked—from human-written counterparts. A common feature of these methods is to examine the complete context, linguistic patterns and other potentially revealing markers in the given text to assess whether it is likely LLM-generated. The simplest method is to build a classifier using synthetic and human text data, which is adopted by some commercial detection platforms [23, 73]. Another category is training-free and leverages the inherent stylistic differences between human and machine writing without specific training data, using techniques such as log probability curvature [6, 43], divergent n-gram analysis [71] and intrinsic dimension estimation [61]. However, [66] find that most post-hoc detection methods are neither accurate nor reliable and suffer from a significant bias toward classifying the output as human-written rather than detecting LLM-generated text. Furthermore, these methods have proven fragile to adversarial attacks and biased against nonnative English writers [33, 38, 52].

2. A statistical framework for watermark detection. In this section, we formally introduce our framework that enables statistical analysis of watermarks. Our focus is on the development of effective techniques for assessing the statistical efficiency of the watermarks accompanying this framework. Henceforth, in this paper, we write $w_{1:n} := w_1 \cdots w_n$ and $\zeta_{1:n} := \zeta_1 \cdots \zeta_n$ for the text and associated pseudorandom variables, respectively.

2.1. *Working hypotheses.* The problem of determining whether a watermark is present or not in the text can be formulated as a hypothesis testing problem [13, 22, 28, 31, 68]:

$$H_0 : w_{1:n} \text{ is written by a human}, \qquad H_1 : w_{1:n} \text{ is written by a watermarked LLM}. \tag{4}$$

In addition to the text, this hypothesis testing problem also uses the pseudorandom variables as data. A unified way to represent ζ_t based on existing constructions [20, 31, 32, 47, 68] is to take the form

$$\zeta_t = \mathcal{A}(w_{1:(t-1)}, \texttt{Key}), \tag{5}$$

where Key denotes a secret key that will be passed to the verifier. The (deterministic) hash function $\mathcal{A}$ maps any token sequence and a key to a pseudorandom number that will be used to generate the next token. More precisely, the watermarked LLM generates the next token according to the rule

$$w_t := \mathcal{S}(\boldsymbol{P}_t, \zeta_t), \tag{6}$$

for some (deterministic) decoder $\mathcal{S}$. Recall that $\boldsymbol{P}$ denotes the probability distribution of the next token generated by the (unwatermarked) LLM.

With these notation in place, a watermark can be formally presented by the tuple $(\mathcal{A}, \mathcal{S}, \texttt{Key})$. To lay a solid footing of watermarks on statistical hypothesis testing, we need two working hypotheses.

WORKING HYPOTHESIS 2.1 (Soundness of pseudorandomness). *In the watermarked LLM, the pseudorandom variables $\zeta_{1:n}$ constructed above are i.i.d. copies of a random variable. Furthermore, ζ_t is (statistically) independent of $w_{1:(t-1)}$.*

Working Hypothesis 2.1 is grounded purely in cryptographic considerations. In cryptography, there are well-established approaches to efficiently constructing and computing the pseudorandom number as a function of text and the secret key [7, 30, 45, 53, 57]. The pseudorandom number is very sensitive to the key, making it computationally *indistinguishable* from its truly random counterpart without knowledge of the key.[7] Specifically, although ζ_t is completely determined by the prior text $w_{1:(t-1)}$ (and the secret key), a run of the hash function $\mathcal{A}$ could effectively introduce fresh randomness, thereby making ζ_t statistically independent of $w_{1:(t-1)}$. Note that, during detection, the communication cost is merely to pass the secret key to the verifier as the prior tokens are publicly available.

Hereafter, we regard ζ as a random variable, which allows us to formally define the *unbiasedness* of a watermark. We say a watermark is unbiased if, for any multinomial distribution $\boldsymbol{P}$, $\mathcal{S}(\boldsymbol{P}, \zeta)$ follows $\boldsymbol{P}$. That is, for any NTP distribution $\boldsymbol{P}$ and token $w \in \mathcal{W}$,

$$\mathbb{P}\big(\mathcal{S}(\boldsymbol{P}, \zeta) = w\big) = P_w,$$

where the expectation is taken over the randomness embodied in ζ. Together with the joint independence of ζ_t's across the sequence of tokens, unbiasedness holds at every step conditional on prior tokens.

Our next working hypothesis concerns the joint distribution of $w_{1:n}$ and $\zeta_{1:n}$ when the text is written by a human.

WORKING HYPOTHESIS 2.2 (Intrinsic nature of human randomness). *Let $w_{1:n}$ be a sequence of tokens generated by a human who has no knowledge of the secret key. Then the human-generated token w_t and ζ_t are (statistically) independent conditional on $(w_{1:(t-1)}, \zeta_{1:(t-1)})$, for all $1 \le t \le n$.*

REMARK 2.1. To clear up any confusion, we remark that the verifier uses the human-generated tokens $w_1, \ldots, w_n$ to compute the random variables $\zeta_{1:n}$ in (5). In particular, Working Hypothesis 2.1 remains valid for human-generated text, due to the construction of the hash function $\mathcal{A}$.

In particular, this working hypothesis shows that, for human-written text, the token is not generated according to (6). The rationale of this working hypothesis is that how a human writes text is intrinsically random and cannot be captured by pseudorandomness. Therefore, the human-generated token w_t has nothing to do with ζ_t. Moreover, one can also argue for this working hypothesis by recognizing that it is practically impossible for a human to generate text such that ζ_t and w_t are dependent because the secret key is not available. This is even the case for a different LLM without having the secret key because it is computationally infeasible to replicate the pseudorandomness without the key.

The two working hypotheses have been adopted by the literature, albeit not as explicit as our treatment. For example, [68, 77] assumed conditions on the pseudorandom hash function so that the first working hypothesis is valid.[8] Other works directly impose distributional assumptions on some summary statistics to effectively satisfy the two working hypotheses

[7] The phrase "computationally indistinguishable" means that no polynomial-time algorithm can distinguish the pseudorandom number from its truly random counterpart without knowledge of the key.

[8] In particular, they assumed that $\mathcal{A}(w_{1:n}, \texttt{Key})$ is i.i.d. for any $w_{1:n}$.

[20, 31, 32, 76]. As a departure from these works, [34] considered a hash function of the form $\zeta_t = \mathcal{A}(t, \texttt{Key})$. Consequently, the independence between the pseudorandom number and prior tokens follows by construction. However, it is worthwhile mentioning that this form of the hash function hampers computational efficiency in detection due to the necessity of iterating through the entire text sequence multiple times.

2.2. *Pivotal statistics.* Understanding the hypothesis testing problem (4) requires analyzing the differences between the joint distribution of $(w_{1:n}, \zeta_{1:n})$ under Working Hypotheses 2.1 and 2.2. This can be seen by first decomposing the joint probability density of $(w_{1:n}, \zeta_{1:n})$ into a product of conditional probabilities:[9]

$$\mathbb{P}(w_{1:n}, \zeta_{1:n}) = \prod_{t=1}^{n} \mathbb{P}(w_t, \zeta_t | w_{1:(t-1)}, \zeta_{1:(t-1)}). \tag{7}$$

- Under H_0, Working Hypotheses 2.1 and 2.2 show that w_t and ζ_t are independent given $(w_{1:(t-1)}, \zeta_{1:(t-1)})$. Hence,[10]

$$\begin{aligned}\mathbb{P}(w_t, \zeta_t | w_{1:(t-1)}, \zeta_{1:(t-1)}) &= \mathbb{P}(w_t | w_{1:(t-1)}, \zeta_{1:(t-1)}) \mathbb{P}(\zeta_t | w_{1:(t-1)}, \zeta_{1:(t-1)}) \\ &= P_{t,w_t} \cdot \mathbb{P}_\zeta(\zeta_t).\end{aligned}$$

  For example, the display above is equal to P_{t,w_t} for the baby watermark defined in (1).
- Under H_1, w_t and ζ_t are dependent given $(w_{1:(t-1)}, \zeta_{1:(t-1)})$. By Working Hypothesis 2.1 and the decoding construction (6), we have

$$\mathbb{P}(w_t, \zeta_t | w_{1:(t-1)}, \zeta_{1:(t-1)}) = \begin{cases} \mathbb{P}_\zeta(\zeta_t) & \text{if } \mathcal{S}(\boldsymbol{P}_t, \zeta_t) = w_t, \\ 0 & \text{if } \mathcal{S}(\boldsymbol{P}_t, \zeta_t) \neq w_t \end{cases}$$

  because the decoder $\mathcal{S}$ is deterministic. For example, the baby watermark satisfies $\mathbb{P}(w_t, \zeta_t | w_{1:(t-1)}, \zeta_{1:(t-1)}) = 1$ if $\zeta_t > P_{t,0}$, $w_t = 1$, or $\zeta_t \leq P_{t,0}$, $w_t = 0$. It is 0 otherwise.

By the Neyman–Pearson lemma, the most powerful test is based on the likelihood ratio:

$$\begin{aligned}\frac{\mathbb{P}_{H_0}(w_{1:n}, \zeta_{1:n})}{\mathbb{P}_{H_1}(w_{1:n}, \zeta_{1:n})} &= \frac{\prod_{t=1}^{n} P_{t,w_t} \cdot \mathbb{P}_\zeta(\zeta_t)}{\prod_{t=1}^{n} \mathbf{1}_{\mathcal{S}(\boldsymbol{P}_t, \zeta_t) = w_t} \cdot \mathbb{P}_\zeta(\zeta_t)} \\ &= \begin{cases} P_{1,w_1} \cdots P_{n,w_n} & \text{if } \mathcal{S}(\boldsymbol{P}_t, \zeta_t) = w_t \text{ for all } t, \\ \infty & \text{otherwise.} \end{cases}\end{aligned} \tag{8}$$

Although the likelihood ratio appears simple, taking only two values, we encounter a significant challenge in distinguishing between the two cases mentioned above. This challenge arises because the NTP distribution $\boldsymbol{P}_t$ is unknown and, worse, can vary with t. This nuisance parameter remains unknown even if the verifier has complete access to the LLM, as the prompt used for generating the text is usually not available to the verifier.

To address this challenge, we seek a pivotal statistic $Y_t = Y(w_t, \zeta_t)$ such that its distribution is the same for any NTP distribution $\boldsymbol{P}_t$ under the null. Such a pivot allows us to construct test statistics for watermark detection with known distributions, and consequently, obtain detection rules with provable Type I error control, though at the price of information

[9] As an abuse of notation, $\mathbb{P}(\cdot)$ is considered the probability density function when applied to a continuous variable, and probability mass function when applied to a discrete variable.

[10] Here, we conceptualize a human as an LLM, using a multinomial distribution to model the selection of the next token based on prior tokens.

loss compared to using the full data, w_t and ζ_t. Formally, the original testing problem (4) is reduced to the following:

$$H_0 : Y_t \sim \mu_0 \quad \text{i.i.d. for } 1 \le t \le n, \qquad H_1 : Y_t \sim \mu_{1,\boldsymbol{P}_t} \text{ for } 1 \le t \le n, \tag{9}$$

where μ_0 denotes the (known) distribution of Y_t when the text is human-written, and $\mu_{1,\boldsymbol{P}_t}$ denotes the (unknown) distribution of Y_t conditional on $(w_{1:(t-1)}, \zeta_{1:(t-1)})$ when the text is generated by the watermarked LLM. As is clear, $\mu_{1,\boldsymbol{P}_t}$ is determined by the NTP distribution $\boldsymbol{P}_t$.

As a caveat, the choice of $Y(w_t, \zeta_t) \equiv \zeta_t$ satisfies pivotality but is useless since the alternative distribution is the same as the null distribution. A useful choice of Y, while being pivotal, should allow the selected token to "pull" the alternative distribution toward the same direction for any NTP distribution $\boldsymbol{P}_t$. For example, $Y(w_t, \zeta_t) = (2w_t - 1)(2\zeta_t - 1)$ is a good choice for the baby watermark since the dependence between w_t and ζ_t (see (1) in Section 1) tends to make $Y(w_t, \zeta_t)$ larger. In general, it is a case-by-case approach to find a reasonable pivot for a given watermark (see Sections 3 and 4).

To test (9), it is natural to use the sum of Y_t's across the token sequence as a test statistic. To enhance the flexibility, we consider a score function h that applies to the pivot Y. This leads to the following rejection rule for the hypothesis testing problem:

$$T_h(Y_{1:n}) := \begin{cases} 1 & \text{if } \sum_{t=1}^{n} h(Y_t) \ge \gamma_{n,\alpha}, \\ 0 & \text{if } \sum_{t=1}^{n} h(Y_t) < \gamma_{n,\alpha}. \end{cases} \tag{10}$$

That is, we reject that the text is written by a human if T_h is above $\gamma_{n,\alpha}$. The threshold $\gamma_{n,\alpha}$ is chosen to ensure significance level at α: $\mathbb{P}_{H_0}(T_h(Y_{1:n}) = 1) = \alpha$. For certain score functions discussed in this work, the distribution of the sum of Y_t's under the null hypothesis admits a closed-form expression, allowing for analytical evaluation of $\gamma_{n,\alpha}$. In general, as the null distribution of $h \circ Y$ is known, an estimator of $\gamma_{n,\alpha}$ when the text is sufficiently long is

$$\hat{\gamma}_{n,\alpha} = n \cdot \mathbb{E}_0 h(Y) + \Phi^{-1}(1-\alpha) \cdot \sqrt{n \cdot \mathrm{Var}_0(h(Y))}, \tag{11}$$

where $\mathbb{E}_0$ and Var_0 indicate that μ_0 is used to take the expectation and variance. As can be seen, the underlying NTP distributions $\boldsymbol{P}_t$'s are not involved in this detection rule.

2.3. *Class-dependent efficiency.* Once the Type I error is controlled, we seek to evaluate the Type II error and use it as a measure to ascertain which choice of score function is more desired than others. If the NTP distributions $\boldsymbol{P}_t$ were known and remained the same with respect to varying t, the optimal score function would simply be given by the log-likelihood ratio, according to the Neyman–Pearson lemma. However, this is not the case. We manage this challenge by assuming that the NTP distributions belong to a distribution class, denoted as $\mathcal{P}$. The flexibility of using a distribution class lies in that one can choose a small class when much is known about the distributional properties of the LLM, and choose a large class if little is known.

Given a distribution class $\mathcal{P}$, we can evaluate the Type II error of the test statistic $T_h(Y_{1:n})$ over the least-favorable NTP distributions in $\mathcal{P}$. This gives rise to a notion of *class-dependent efficiency*. For any score function h and NTP distribution $\boldsymbol{P}$, we define the following moment-generating function (MGF):

$$\phi_{\boldsymbol{P},h}(\theta) := \mathbb{E}_{1,\boldsymbol{P}} e^{-\theta h(Y)}, \tag{11}$$

where $\mathbb{E}_{1,\boldsymbol{P}}$ indicates that the expectation is taken over the randomness embodied in $Y \sim \mu_{1,\boldsymbol{P}}$ in (9). Assuming that the NTP distributions are all in $\mathcal{P}$, the following result delineates the Type II error in the large sample limit. We defer its proof to Supplement A, which relies on techniques in large deviation theory [17, 62].

THEOREM 2.1. *Assume $\boldsymbol{P}_t \in \mathcal{P}$ for all t. For any h satisfying $\mathbb{E}_0|h| < \infty$, the Type II error of the detection rule T_h defined in (10) obeys*

$$\limsup_{n\to\infty} \mathbb{P}_{H_1}\big(T_h(Y_{1:n}) = 0\big)^{1/n} \le \mathrm{e}^{-R_{\mathcal{P}}(h)}, \tag{12}$$

where $R_{\mathcal{P}}(h)$ is given by

$$R_{\mathcal{P}}(h) = -\inf_{\theta\ge 0} \sup_{\boldsymbol{P}\in\mathcal{P}} \big(\theta\mathbb{E}_0 h(Y) + \log \phi_{\boldsymbol{P},h}(\theta)\big) = -\inf_{\theta\ge 0}\Big(\theta\mathbb{E}_0 h(Y) + \sup_{\boldsymbol{P}\in\mathcal{P}} \log \phi_{\boldsymbol{P},h}(\theta)\Big). \tag{13}$$

REMARK 2.2. $R_{\mathcal{P}}(h)$ is the $\mathcal{P}$-dependent efficiency rate of h, or simply the $\mathcal{P}$-efficiency rate.

REMARK 2.3. Under a certain regularity condition, (12) is tight in the sense that there exists $\boldsymbol{P}^\star$ in $\mathcal{P}$ such that, if $\boldsymbol{P}_t = \boldsymbol{P}^\star$ for all t, then $\mathbb{P}_{H_1}(T_h(Y_{1:n}) = 0) \ge \mathrm{e}^{-(R_{\mathcal{P}}(h)+\varepsilon)n}$ for any positive ε and sufficiently large n. The regularity condition is satisfied by both the Gumbel-max and inverse transform watermarks and is detailed in the remark following the proof of Theorem 2.1 in Supplement A.

Proposition 2.1 shows the Type II error decays exponentially as long as the exponential component $R_{\mathcal{P}}(h)$ is positive. The larger the value of $R_{\mathcal{P}}(h)$ is, the more efficient the detection rule T_h is. Put differently, this class-dependent measure of efficiency reduces comparing watermark detection rules to the rate of class-dependent efficiency.

This efficiency rate, obtained from large deviation theory, does not depend on the specific value of α. The same is true for Bahadur efficiency [5], which quantifies the rate at which the significance level (p-value) of a test statistic approaches zero with increasing sample size. It emphasizes the control of Type I errors for simple hypothesis testing with i.i.d. data. In contrast, our metric $R_{\mathcal{P}}(h)$ centers on quantifying the least decline of Type II errors to zero when the underlying NTP distribution $\boldsymbol{P}$ is picked up from the distribution class $\mathcal{P}$. In short, it focuses on composite hypothesis testing where the observed data is not identically distributed.

Optimality via minimax optimization. More importantly, the notion of class-dependent efficiency serves as a concrete approach to identifying the optimal score function that achieves the largest possible value of $\mathcal{P}$-efficiency rate. Following from (13), formally, this amounts to solving the following optimization problem:

$$\sup_h R_{\mathcal{P}}(h) = -\inf_{h,\theta\ge 0} \sup_{\boldsymbol{P}\in\mathcal{P}} \big(\mathbb{E}_0\theta h(Y) + \log \mathbb{E}_{1,\boldsymbol{P}}\mathrm{e}^{-\theta h(Y)}\big).$$

By viewing θh as a new score function, finding the optimal h is reduced to solving the following minimax optimization program:[11]

$$\min_h \max_{\boldsymbol{P}\in\mathcal{P}} L(h, \boldsymbol{P}) \quad \text{where } L(h, \boldsymbol{P}) := \mathbb{E}_0 h(Y) + \log \mathbb{E}_{1,\boldsymbol{P}}\mathrm{e}^{-h(Y)}. \tag{14}$$

The function $L(h, \boldsymbol{P})$ is convex in the score function h for any fixed $\boldsymbol{P}$, but is generally not concave in $\boldsymbol{P}$ when h is fixed. Therefore, this minimax optimization problem is generally not convex–concave, making it challenging to solve (14) numerically [39, 51].

[11] To highly the minimax nature of this formulation, we use max and min in place of sup and inf, respectively.

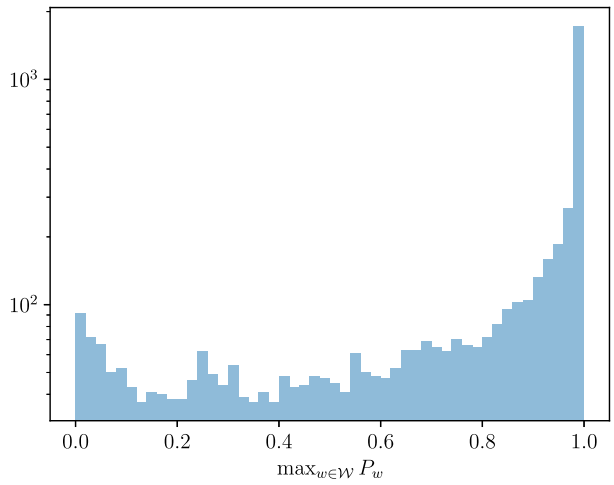


FIG. 1. *Empirical frequency of* $\max_{w\in\mathcal{W}} P_{t,w}$ *using outputs from ChatGPT-3.5-turbo. There are a total of* 4997 *top-one probabilities recorded,* 81.36% *of which are less than* 0.999 *while* 99.79% *are larger than* 0.001. *See Supplement D.3 for the experimental setup.*

Interestingly, we will show in Sections 3 and 4 that, for the Gumbel-max and inverse transform watermarks, the minimax optimization problem possesses certain structural properties that enable us to *analytically* identify global solutions to (14).

Choice of distribution classes. The choice of the distribution class $\mathcal{P}$ is crucial since it follows from (13) that $R_{\mathcal{P}_1}(h) \geq R_{\mathcal{P}_2}(h)$ if $\mathcal{P}_1 \subset \mathcal{P}_2$. While it might be plausible to assume a fraction of NTP distributions are in a "nice" class, some might be outside. The following result extends Proposition 2.1 to this practical scenario.

PROPOSITION 2.1. *Assume that at least* γ*-fraction of* $\boldsymbol{P}_1, \ldots, \boldsymbol{P}_n$ *is in* $\mathcal{P}_1$ *with the rest being in* $\mathcal{P}_2$,[12] *where* $0 < \gamma < 1$ *and* $\mathcal{P}_1 \subset \mathcal{P}_2$. *Then, for any* h, *we have*

$$\limsup_{n\to\infty} \mathbb{P}_{H_1}\big(T_h(Y_{1:n}) = 0\big)^{1/n} \leq \mathrm{e}^{-\{\gamma\cdot R_{\mathcal{P}_1}(h) + (1-\gamma)\cdot R_{\mathcal{P}_2}(h)\}}.$$

When there is no prior at all about the LLMs, the associated class is the $(|\mathcal{W}| - 1)$-dimensional simplex, which contains all possible NTP distributions and we denote by $\mathrm{Simp}(\mathcal{W})$. This distribution class includes *singular* distributions—a distribution $\boldsymbol{P}$ such that $P_w = 1$ for some token w—for which the pivotal statistic has the same distribution under the null and alternative in (9). Formally, this gives $R_{\mathrm{Simp}(\mathcal{W})}(h) = 0$, that is, the class-dependent efficiency of $\mathrm{Simp}(\mathcal{W})$ is zero. Taking $\mathcal{P}_1 = \mathcal{P} \subset \mathcal{P}_2 = \mathrm{Simp}(\mathcal{W})$, Proposition 2.1 shows $\limsup_{n\to\infty} \mathbb{P}_{H_1}(T_h(Y_{1:n}) = 0)^{1/n} \leq \mathrm{e}^{-\gamma R_{\mathcal{P}}(h)}$. This result implies that when only a fraction of NTP distributions is known to some extent, the Type II error is still governed by $R_{\mathcal{P}}(h)$ up to a multiplicative constant γ. The comparison between choices of the score functions h can still be reduced to comparing the value of $R_{\mathcal{P}}(h)$.

In addition, the discussion above reveals that any detection rule would become powerless if nothing is known about the NTP distributions. This justifies the necessity of imposing structural assumptions on the distribution class. For $0 < \Delta \leq 1 - \frac{1}{|\mathcal{W}|}$, we call

$$\mathcal{P}_\Delta := \Big\{\boldsymbol{P} : \max_{w\in\mathcal{W}} P_w \leq 1 - \Delta\Big\}$$

[12]We do not need to know which are in the γ fraction.

the Δ-regular distribution class. Accordingly, an NTP distribution $\boldsymbol{P}$ is called Δ-regular if $\boldsymbol{P} \in \mathcal{P}_\Delta$. This excludes singular distributions. Interestingly, Δ-regularity is closely related to the Shannon entropy. For any Δ-regular NTP distribution $\boldsymbol{P}$, its Shannon entropy satisfies $\mathrm{Ent}(\boldsymbol{P}) = \sum P_w \log \frac{1}{P_w} \geq \sum P_w(1 - P_w) \geq \sum P_w \cdot \Delta = \Delta$, where the first inequality follows because $\log \frac{1}{1-x} \geq x$ for $x < 1$. This shows that Δ-regularity imposes a lower bound on how much information the NTP distribution offers. In practice, most NTP distributions are Δ-regular for a proper value of Δ. See Figure 1 for an empirical investigation where 88.65% of the NTP distributions are 0.0001-regular, 81.36% are 0.001-regular and 70.28% are 0.01-regular.

This distribution class offers a way to measure the performance of a detection rule using the class-dependent efficiency in (13). For example, the $\mathcal{P}_\Delta$-efficiency rate of the baby watermark using the statistic (2) is

$$-\inf_{\theta \geq 0} \log\left[\frac{1}{\theta}\left\{\frac{\mathrm{e}^{\theta(1-2\Delta)} + \mathrm{e}^{-\theta(1-2\Delta)}}{2} - \mathrm{e}^{-\theta}\right\}\right].$$

A proof of this fact is deferred to Supplement B.8.

In practice, Δ-regularity is closely related to the temperature parameter in LLMs [3], which is used to divide the raw output of a model's last layer by a temperature parameter before applying the softmax function. A high temperature leads to more uniform probabilities (encouraging exploration), while a low temperature makes the distribution sharper, emphasizing the most likely outcomes (encouraging exploitation).

3. Application to the Gumbel-max watermark. In this section, we apply the framework to the Gumbel-max watermark [1]. Recall that the Gumbel-max decoder can also be written as

$$w_t = \mathcal{S}^{\mathrm{gum}}(\boldsymbol{P}_t, \zeta_t) := \arg\max_{w \in \mathcal{W}} V^{\mathrm{gum}}(P_{t,w}, U_{t,w}) \quad \text{where } V^{\mathrm{gum}}(p, u) := \frac{\log u}{p}. \tag{15}$$

Above, $\{\zeta_t\}_{t=1}^n = \{(U_{t,w})_{w \in \mathcal{W}}\}_{t=1}^n$ represents $n \times |\mathcal{W}|$ i.i.d. replicates of the standard uniform random variable $U(0, 1)$. As seen from (15), the Gumbel-max trick ensures that this decoder is unbiased for sampling from the NTP distribution $\boldsymbol{P}_t$. In implementation, ζ_t can be computed using a pseudorandom hash function $\mathcal{A}$ that depends only on, for example, the last five tokens $w_{t-5}, \ldots, w_{t-1}$ in addition to the secrete key [1].

Uniqueness of the Gumbel-max decoder. As a deviation from the main focus, we are tempted to ask whether other forms of Gumbel-max-style decoders remain unbiased. Consider an unbiased decoder that takes the form

$$\mathcal{S}^V(\boldsymbol{P}, \zeta) := \arg\max_{w \in \mathcal{W}} V(P_w, U_w)$$

for some function V. When the vocabulary size $|\mathcal{W}| = 2$, writing $\boldsymbol{P} = (P_0, P_1)$, unbiasedness requires[13]

$$\mathbb{P}\big(\mathcal{S}^V(\boldsymbol{P}, \zeta) = 0\big) = \mathbb{P}\big(V(P_0, U_0) \geq V(P_1, U_1)\big) = \frac{P_0}{P_0 + P_1}. \tag{16}$$

When there are more than two tokens in the vocabulary, a natural requirement is that, by "gluing" two tokens into a new token, the decoder remains unbiased for the new vocabulary that has been reduced by one in size. This new token is sampled whenever either of the two

[13] In (16), it evaluates to P_0 since $P_0 + P_1 = 1$. But we regard (16) as an identity for any P_0 and P_1 such that $P_0 + P_1 \leq 1$.

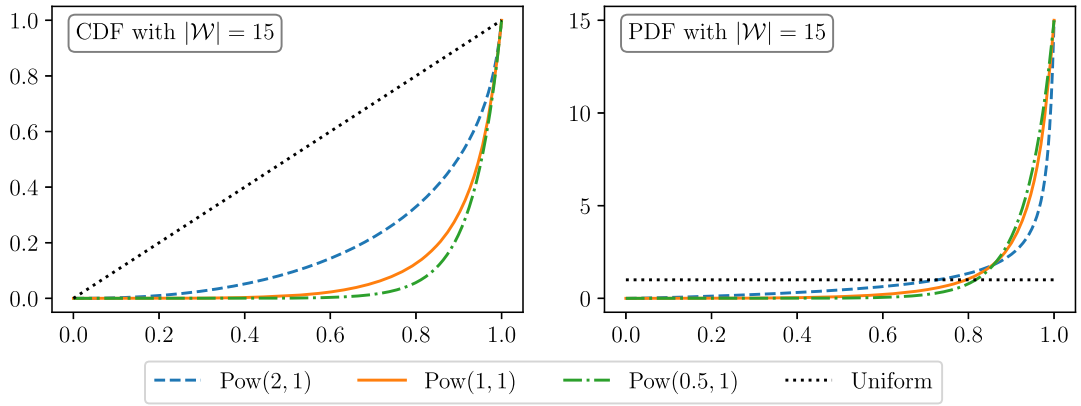


FIG. 2. *Illustration of the distribution of* Y_t^{gum}. *We plot the CDF* $F_{1,\boldsymbol{P}_t}$ *and the PDF* $f_{\boldsymbol{P}_t}$ *by assuming* $P_{t,w} \propto (w+b)^{-a}$, $w \in \mathcal{W}$ *with* $|\mathcal{W}| = 15$ *and different choices of* (a, b). *The resulting distribution is denoted by* $\text{Pow}(a, b)$. $F_{1,\boldsymbol{P}_t}$ *deviates from the uniform distribution.*

original tokens is sampled, which occurs with probability $P_0 + P_1$. This would be true if we have

$$\max\{V(P_0, U_0), V(P_1, U_1)\} \overset{d}{=} V(P_0 + P_1, U), \tag{17}$$

where $\overset{d}{=}$ denotes equality in distribution and $U \sim U(0, 1)$.

Both (16) and (17) are satisfied by Aaronson's choice V^{gum} for any P_0 and P_1 such that $P_0 + P_1 \le 1$. Interestingly, our following theorem shows that this choice is essentially unique. Its proof is deferred to Supplement B.2.

THEOREM 3.1. *Suppose* V *satisfies both* (16) *and* (17). $\mathcal{S}^V$ *is unbiased if and only if*

$$V(p, u) = g\left(\frac{\log u}{p}\right) \equiv g\big(V^{\text{gum}}(p, u)\big)$$

for a strictly increasing function g.

3.1. *Main results.*

Optimal score function. Our framework starts by identifying a pivot for the hypothesis testing problem (4). The random number U_{t,w_t}, which corresponds to the selected token w_t at step t, is a good candidate. This is because U_{t,w_t} follows the standard uniform distribution under H_0 while being stochastically larger under H_1, as ensured by Working Hypotheses 2.1 and 2.2. Indeed, U_{t,w_t} is used in [1] for its detection rule.

We write $Y_t^{\text{gum}} \equiv Y^{\text{gum}}(w_t, \zeta_t) := U_{t,w_t}$. This choice of pivot can be seen as an approximation of the most "optimistic" log-likelihood ratio used in an independent hypothesis test for watermark detection (see Supplement B.1 for details). The distribution of this pivot under H_1 is a mixture of Beta distributions, which is a classic result in the literature.

LEMMA 3.1 ([20, 47]). *Under* H_1, *the distribution of* Y_t^{gum} *given* $\boldsymbol{P}_t$ *obeys*

$$\mathbb{P}_{H_1}\big(Y_t^{\text{gum}} \le r | \boldsymbol{P}_t\big) = \sum_{w \in \mathcal{W}} P_{t,w} r^{1/P_{t,w}} \quad \text{for } r \in [0, 1].$$

The fact that this distribution is stochastically larger than $U(0, 1)$ can be gleaned from the following inequality: $\sum_{w \in \mathcal{W}} P_{t,w} r^{1/P_{t,w}} = r \sum_{w \in \mathcal{W}} P_{t,w} r^{1/P_{t,w}-1} \le r \sum_{w \in \mathcal{W}} P_{t,w} = r$.

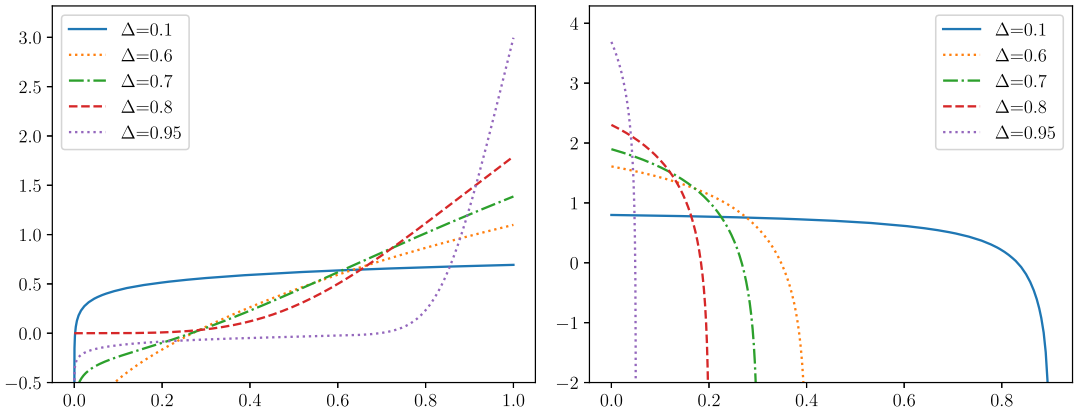


FIG. 3. *Illustration of* $h^\star_{\mathrm{gum},\Delta}$ *(left) and* $h^\star_{\mathrm{dif},\Delta}$ *(right) for different values of* Δ.

Also, see Figure 2 for an illustration: there is a higher accumulation of probability mass around one under H_1 compared to the distribution observed under H_0. From this observation, any score function h aware of the distributional differences could be used for detecting the Gumbel-max watermarks. For example, [1] uses $h_{\mathrm{ars}}(r) = -\log(1-r)$, while [20, 34] propose $h_{\log}(r) = \log r$. All of these score functions are increasing functions that assign large values to points near one. While all these choices control the Type I error, a crucial question is on what basis we compare score functions and which is the optimal one.

To proceed under the framework, we consider the class-dependent efficiency with distribution class $\mathcal{P}_\Delta$ for $\Delta \in [0, 1 - |\mathcal{W}|^{-1}]$. As one of the main findings of this paper, we obtain the score function with the highest $\mathcal{P}_\Delta$-efficiency rate. Its proof constitutes the subject of Section 3.2. Below, $\lfloor x \rfloor$ denotes the greatest integer that is less than or equal to x.

THEOREM 3.2. *Let* $h^\star_{\mathrm{gum},\Delta} : [0,1] \to \mathbb{R}$ *be defined as*

$$h^\star_{\mathrm{gum},\Delta}(r) = \log\left(\left\lfloor \frac{1}{1-\Delta} \right\rfloor r^{\frac{\Delta}{1-\Delta}} + r^{\frac{\widetilde{\Delta}}{1-\Delta}}\right) \quad \text{with } \widetilde{\Delta} = (1-\Delta)\left\lfloor \frac{1}{1-\Delta} \right\rfloor. \tag{18}$$

This function gives the optimal $\mathcal{P}_\Delta$*-efficiency rates in the sense that* $R_{\mathcal{P}_\Delta}(h^\star_{\mathrm{gum},\Delta}) \geq R_{\mathcal{P}_\Delta}(h)$ *for any measurable function* h*. Moreover,* $R_{\mathcal{P}_\Delta}(h^\star_{\mathrm{gum},\Delta})$ *is attained at the following least-favorable NTP distribution in* $\mathcal{P}_\Delta$:

$$\boldsymbol{P}^\star_\Delta = \Bigg(\underbrace{1-\Delta, \ldots, 1-\Delta}_{\lfloor \frac{1}{1-\Delta} \rfloor \text{ times}}, 1 - (1-\Delta)\cdot\left\lfloor \frac{1}{1-\Delta} \right\rfloor, 0, \ldots\Bigg). \tag{19}$$

REMARK 3.1. To clear up any confusion, the second conclusion amounts to saying that $(h^\star_{\mathrm{gum},\Delta}, \boldsymbol{P}^\star_\Delta)$ is a saddle point solution to the minimax problem (14). Explicitly, $\boldsymbol{P}$ maximizes $L(h^\star_{\mathrm{gum},\Delta}, \boldsymbol{P})$ in (14) if and only if $\boldsymbol{P}$ results from rearranging $\boldsymbol{P}^\star_\Delta$'s coordinates.

The significance of Theorem 3.2 lies partly in the closed-form nature of the optimal score function, thereby making it easy to use in detecting LLM-generated text. The left panel of Figure 3 shows $h^\star_{\mathrm{gum},\Delta}$ for some values of Δ. By the Neyman–Pearson lemma, the optimal score function and least-favorable distribution are related via $h^\star_{\mathrm{gum},\Delta} = \log(\mathrm{d}\mu_{1,\boldsymbol{P}}/\mathrm{d}\mu_0)|_{\boldsymbol{P}=\boldsymbol{P}^\star_\Delta}$. The left panel of Figure 4 shows the efficiency $R_{\mathcal{P}_\Delta}(h^\star_{\mathrm{gum},\Delta})$ as a function of Δ. This function is not smooth when $\Delta = \frac{k}{k+1}$ for integer $k \geq 1$ because the support of $\boldsymbol{P}^\star_\Delta$ jumps at these values.

As an interesting fact, the least-favorable distribution $\boldsymbol{P}^{\star}_{\Delta}$ and its permuted counterparts form *all* the vertices of $\mathcal{P}_{\Delta}$ (see Lemma 3.4 in Section 3.2). Moreover, the vertices are the closest NTP distributions in $\mathcal{P}_{\Delta}$ to singular distributions. Roughly speaking, the closer to singular distributions, the more difficult it is to tease apart the two hypotheses in (9). To see this, note that the testing problem becomes most difficult when the NTP distribution is singular, which makes the null and alternative distributions of the pivot indistinguishable.

Comparison with other rules. While it is now known that $h^{\star}_{\mathrm{gum},\Delta}$ is optimal in the sense of $\mathcal{P}_{\Delta}$-efficiency, it remains interesting to analyze the relative efficiency of other existing rules. In addition to the aforementioned h_{ars} and $h_{\log}$, we consider $h_{\mathrm{ind},\delta}(r) = \mathbf{1}_{\{r \geq \delta\}}$ for a threshold $0 < \delta < 1$, say, $\delta = \mathrm{e}^{-1}$.[14] This indicator function is included as it is perhaps the simplest score function detecting the distributional differences between H_0 and H_1.

By making use of Theorem 2.1 within our framework, we compare these four detection rules in terms of $\mathcal{P}_{\Delta}$-efficiency in the theorem below.

THEOREM 3.3. *There exists an absolute constant* $\Delta^{\star} \approx 0.17756$ *such that the following two statements hold*:

(a) *When* $0.001 < \Delta < \Delta^{\star}$, h_{ars} *has higher* $\mathcal{P}_{\Delta}$*-efficiency than both* $h_{\log}$ *and* $h_{\mathrm{ind},\mathrm{e}^{-1}}$, *that is*,

$$\max\{R_{\mathcal{P}_{\Delta}}(h_{\log}), R_{\mathcal{P}_{\Delta}}(h_{\mathrm{ind},\mathrm{e}^{-1}})\} < R_{\mathcal{P}_{\Delta}}(h_{\mathrm{ars}}) < R_{\mathcal{P}_{\Delta}}(h^{\star}_{\mathrm{gum},\Delta}).$$

(b) *When* $\Delta^{\star} < \Delta < 0.99$, $h_{\log}$ *has higher* $\mathcal{P}_{\Delta}$*-efficiency than both* h_{ars} *and* $h_{\mathrm{ind},\mathrm{e}^{-1}}$, *that is*,

$$\max\{R_{\mathcal{P}_{\Delta}}(h_{\mathrm{ars}}), R_{\mathcal{P}_{\Delta}}(h_{\mathrm{ind},\mathrm{e}^{-1}})\} < R_{\mathcal{P}_{\Delta}}(h_{\log}) < R_{\mathcal{P}_{\Delta}}(h^{\star}_{\mathrm{gum},\Delta}).$$

REMARK 3.2. The constant $\Delta^{\star}$ is the unique root of the equation $R_{\mathcal{P}_{\Delta}}(h_{\mathrm{ars}}) = R_{\mathcal{P}_{\Delta}}(h_{\log})$ for $\Delta > 0.001$. Numerical evaluation shows $\Delta^{\star} = 0.17756080525\ldots$. A proof of Theorem 3.3 can be found in Supplement B.6.

Figure 4 illustrates Theorem 3.3 by presenting the comparisons under our statistical framework. Notably, for small values of Δ, $\max_{w \in \mathcal{W}} P_{t,w}$ can be close to 1. In this regime, $\mu_{1,\boldsymbol{P}_t}$ is close to μ_0 in distribution, making the detection efficiency relatively low regardless of the

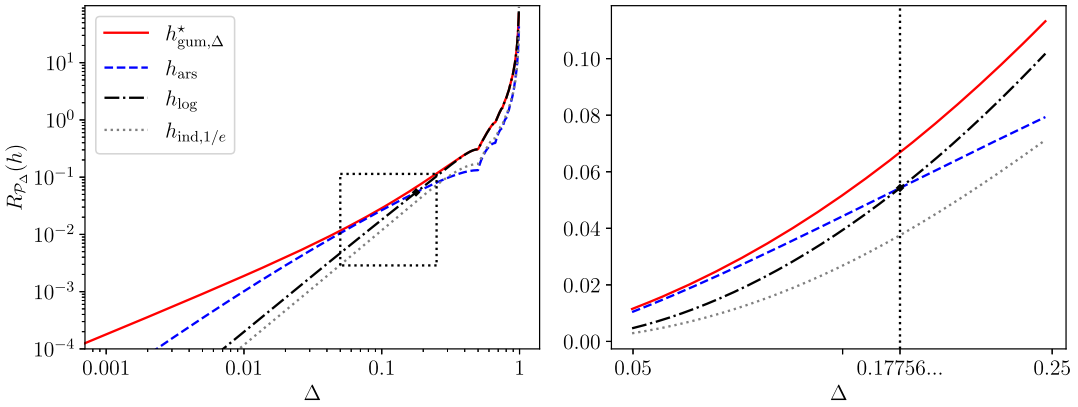


FIG. 4. *$\mathcal{P}_{\Delta}$-efficiency rates of detection methods for Gumbel-max watermarks. Note that $R_{\mathcal{P}_{\Delta}}(h)$ has nonsmooth points when $\Delta = \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$.*

[14] According to the $\overline{R}_{\Delta}$-efficiency defined in (25), $\delta = \mathrm{e}^{-1}$ achieves the largest $\overline{R}_{\Delta}$-efficiency when $\Delta \to 1$. See Lemma C.7 for details.

rule used. Conversely, a large value of Δ implies all the token probabilities tend to be small, leading to a significant discrepancy between $\mu_{1,\boldsymbol{P}_t}$ and μ_0. Accordingly, all detection rules have efficiency rates tending to infinity.

While $h^{\star}_{\mathrm{gum},\Delta}$ achieves the highest $\mathcal{P}_\Delta$-efficiency rate across the entire range of Δ, the relative performance of h_{ars} and $h_{\log}$ depends on the value of Δ. From an empirical perspective, h_{ars} is observed to outperform $h_{\log}$ in common settings of LLMs [20, 34]. This is because the largest token probability in NTP distributions is, by and large, close to 1. To appreciate this fact, we experiment on ChatGPT-3.5-turbo with twenty prompts (see Supplement D.3 for details) and track the largest token probabilities across the generation of token sequences. The results are presented in Figure 1, which shows that, for example, 56.85% of the NTP distributions have the largest token probability above $1-\Delta^{\star}\approx 0.82$. This is the regime where h_{ars} is superior to $h_{\log}$, as predicted by our Theorem 3.3.

Nevertheless, Δ can be large when the LLM generation is highly stochastic. In this regime, $h_{\mathrm{ind},\mathrm{e}^{-1}}$ slightly outperforms h_{ars}, but the gap soon diminishes to zero as Δ tends to 1, as seen from Figure 4. Similarly, $h_{\log}$ has a diminishing suboptimality compared with the optimal $h^{\star}_{\mathrm{gum},\Delta}$ in the sense that $\lim_{\Delta\to 1} R_{\mathcal{P}_\Delta}(h_{\log})/R_{\mathcal{P}_\Delta}(h^{\star}_{\mathrm{gum},\Delta})\to 1$. We prove the limiting behaviors of these score functions in Supplement B.7.

3.2. *Proof of Theorem 3.2.* The proof of Theorem 3.2 relies on the minimax formulation (14) in our framework to solve for the optimal score function. As mentioned earlier, however, this task is generally difficult since $L(h,\boldsymbol{P})=\mathbb{E}_0 h(Y)+\log\mathbb{E}_{1,\boldsymbol{P}}\mathrm{e}^{-h(Y)}$ is not convex–concave. We circumvent this obstacle by introducing a simple yet remarkable result that characterizes the MGF (11) associated with the Gumbel-max watermark.

LEMMA 3.2 (Convexity lemma). *For any nondecreasing function h, the functional*[15] $\boldsymbol{P}\mapsto\phi_h(\boldsymbol{P}):=\mathbb{E}_{1,\boldsymbol{P}}\mathrm{e}^{-h(Y^{\mathrm{gum}})}$ *is convex in* $\boldsymbol{P}\in\mathrm{Simp}(\mathcal{W})$.

This convexity lemma is the key to the proof of Theorem 3.2 and is a contribution of independent interest to future work on LLM watermarks. Roughly speaking, it helps reduce the "max" part in the minimax problem (14) to the problem of identifying vertices of the distribution class. Technically speaking, the role of the convexity lemma in the proof of Theorem 3.2 is through the following lemma.

LEMMA 3.3. *For any nondecreasing function h, we have*

$$\sup_{\boldsymbol{P}\in\mathcal{P}_\Delta}\phi_h(\boldsymbol{P})=\phi_h\big(\boldsymbol{P}^{\star}_{\Delta}\big). \tag{20}$$

REMARK 3.3. A direct consequence of Lemma 3.3 is that $R_{\mathcal{P}_\Delta}(h)=-\inf_{\theta\ge 0}L(\theta h,\boldsymbol{P}^{\star}_{\Delta})$.

Now we prove Lemmas 3.2 and 3.3 in order.

PROOF OF LEMMA 3.2. By Lemma 3.1, the CDF of Y^{gum} with NTP distribution $\boldsymbol{P}$ takes the form: $F_{1,\boldsymbol{P}}(r)=\sum_{w\in\mathcal{W}}P_w r^{1/P_w}$ for $r\in[0,1]$. This function relates to $\phi_h(\boldsymbol{P})$ since

$$\phi_h(\boldsymbol{P})=\mathbb{E}_{1,\boldsymbol{P}}\mathrm{e}^{-h(Y^{\mathrm{gum}})}=\int_0^1\mathrm{e}^{-h(r)}F_{1,\boldsymbol{P}}(\mathrm{d}r).$$

[15] We consider a slightly different version of the MGF by setting $\theta=1$ and taking as input $\boldsymbol{P}$ in (11).

First, we show that $\boldsymbol{P} \mapsto F_{1,\boldsymbol{P}}(r)$ is a convex function for any given $r \in [0,1]$. This is demonstrated by showing that the Hessian matrix of $\boldsymbol{P} \mapsto F_{1,\boldsymbol{P}}(r)$ is positive semidefinite for any $\boldsymbol{P}$ within the interior of its domain $\mathrm{Simp}(\mathcal{W})$:

$$\nabla^2_{\boldsymbol{P}} F_{1,\boldsymbol{P}}(r) = \begin{bmatrix} r^{1/P_1}\dfrac{\log^2 r}{P_1^3} & 0 & \cdots & 0 \\ 0 & r^{1/P_2}\dfrac{\log^2 r}{P_2^3} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & r^{1/P_{|\mathcal{W}|}}\dfrac{\log^2 r}{P_{|\mathcal{W}|}^3} \end{bmatrix} \succeq 0.$$

Note the Hessian is diagonal with all nonnegative entries, confirming the convexity of $\boldsymbol{P} \mapsto F_{1,\boldsymbol{P}}(r)$.

Second, we examine the functional form of $\phi_h(\boldsymbol{P})$ through integration by parts, which yields

$$\phi_h(\boldsymbol{P}) = F_{1,\boldsymbol{P}}(r)\mathrm{e}^{-h(r)}|_0^1 + \int_0^1 F_{1,\boldsymbol{P}}(r)\mathrm{e}^{-h(r)}h(\mathrm{d}r) = \mathrm{e}^{-h(1)} + \int_0^1 F_{1,\boldsymbol{P}}(r)\mathrm{e}^{-h(r)}h(\mathrm{d}r).$$

This equation expresses $\phi_h(\boldsymbol{P})$ as a nonnegative weighted sum of $F_{1,\boldsymbol{P}}(r)$ evaluated over $r \in [0,1]$.

Given the established convexity of $\boldsymbol{P} \mapsto F_{1,\boldsymbol{P}}(r)$, and considering that a nonnegative weighted sum of convex functions remains convex, it follows directly that $\boldsymbol{P} \mapsto \phi_h(\boldsymbol{P})$ is convex. This completes the proof of Lemma 3.2. □

PROOF OF LEMMA 3.3. Our proof is based on a fundamental principle in convex analysis: the supremum of a convex function over a compact convex set in a Euclidean space is necessarily attained at an extreme point of the convex set [26].

Fix a nondecreasing function h. The mapping $\boldsymbol{P} \mapsto \phi_h(\boldsymbol{P})$ is shown to be convex in $\boldsymbol{P}$ by Lemma 3.3. Additionally, $\mathcal{P}_\Delta$ is convex. Thus, the supremum of the convex function $\boldsymbol{P} \mapsto \phi_h(\boldsymbol{P})$ must be necessarily attained at an extreme point of the convex constraint set $\mathcal{P}_\Delta$. By Lemma 3.4, any extreme point of the set $\mathcal{P}_\Delta$ must be a permuted $\boldsymbol{P}^\star_\Delta$.

LEMMA 3.4. *The set of extreme points of* $\mathcal{P}_\Delta$, *denoted by* $\mathrm{Ext}(\mathcal{P}_\Delta)$, *is given by*

$$\mathrm{Ext}(\mathcal{P}_\Delta) = \{\pi(\boldsymbol{P}^\star_\Delta) : \pi \text{ is a permutation on } \{1, 2, \ldots, |\mathcal{W}|\}\},$$

where $\pi(\boldsymbol{P})$ *denote the permuted NTP distribution whose* i*th coordinate is* $P_{\pi(i)}$.

Note there is a permutation invariance in the objective: $\phi_h(\boldsymbol{P}) = \phi_h(\pi(\boldsymbol{P}))$ holds for every permutation π on $\{1, 2, \ldots, |\mathcal{W}|\}$ and every vector $\boldsymbol{P}$. As a consequence, we can explicitly compute the supremum of $\phi_h(\boldsymbol{P})$ on the convex set $\mathcal{P}_\Delta$ as follows:

$$\sup_{\boldsymbol{P}\in\mathcal{P}_\Delta} \phi_h(\boldsymbol{P}) = \sup_{\boldsymbol{P}\in\mathrm{Ext}(\mathcal{P}_\Delta)} \phi_h(\boldsymbol{P}) = \phi_h(\boldsymbol{P}^\star_\Delta).$$

This proves equation (20). The remaining statement in Lemma 3.3 then follows. □

We conclude this section by proving Theorem 3.2.

PROOF OF THEOREM 3.2. The second part of this theorem follows directly from the convexity lemma (Lemma 3.2) and Lemma 3.4.

To prove the first part, we work on the minimax formulation (14) and begin by recognizing that

$$\begin{aligned}\min_h \max_{\boldsymbol{P}\in\mathcal{P}_\Delta} L(h,\boldsymbol{P}) &= \min_h \max_{\boldsymbol{P}\in\mathcal{P}_\Delta} \big(\mathbb{E}_0 h\big(Y^{\mathrm{gum}}\big) + \log \phi_h(\boldsymbol{P})\big) \\ &\geq \min_h \big(\mathbb{E}_0 h\big(Y^{\mathrm{gum}}\big) + \log \phi_h\big(\boldsymbol{P}^\star_\Delta\big)\big) = -D_{\mathrm{KL}}(\mu_0, \mu_{1,\boldsymbol{P}^\star_\Delta}),\end{aligned} \tag{21}$$

where the final equality is due to the Donsker–Varadhan representation [19] and D_{KL} denotes the Kullback–Leibler divergence. Note that Lemma 3.3 cannot be applied here because the minimization function h is not necessarily nondecreasing.

Due to the uniqueness of the Donsker–Varadhan representation, $\mathbb{E}_0 h(Y^{\mathrm{gum}}) + \log \phi_h(\boldsymbol{P}^\star_\Delta)$ is strictly larger than $-D_{\mathrm{KL}}(\mu_0, \mu_{1,\boldsymbol{P}^\star_\Delta})$ unless we take the log-likelihood ratio

$$h^\star_{\mathrm{gum},\Delta}(r) = \log \frac{\mathrm{d}\mu_{1,\boldsymbol{P}^\star_\Delta}}{\mathrm{d}\mu_0}(r) = \log\left(\left\lfloor \frac{1}{1-\Delta} \right\rfloor r^{\frac{\Delta}{1-\Delta}} + r^{\frac{\tilde{\Delta}}{1-\tilde{\Delta}}}\right),$$

which is nondecreasing in r. In this case, we get

$$\begin{aligned}\max_{\boldsymbol{P}\in\mathcal{P}_\Delta} L\big(h^\star_{\mathrm{gum},\Delta}, \boldsymbol{P}\big) &= \max_{\boldsymbol{P}\in\mathcal{P}_\Delta} \big(\mathbb{E}_0 h^\star_{\mathrm{gum},\Delta}\big(Y^{\mathrm{gum}}\big) + \log \phi_{h^\star_{\mathrm{gum},\Delta}}(\boldsymbol{P})\big) \\ &= \mathbb{E}_0 h^\star_{\mathrm{gum},\Delta}\big(Y^{\mathrm{gum}}\big) + \log \phi_{h^\star_{\mathrm{gum},\Delta}}\big(\boldsymbol{P}^\star_\Delta\big) = -D_{\mathrm{KL}}(\mu_0, \mu_{1,\boldsymbol{P}^\star_\Delta}),\end{aligned} \tag{22}$$

where the final equality follows from Lemma 3.3 and the third equality is ensured by, again, the Donsker–Varadhan representation.

Taken together, (21) and (22) show that $h^\star_{\mathrm{gum},\Delta}$ is the unique score function that solves the minimax problem (14) with $\boldsymbol{P}^\star_\Delta$. □

4. Application to the inverse transform watermark. In this section, we apply the framework to the inverse transform watermark [34]. Without loss of generality, below we take $\mathcal{W} = \{1, \ldots, |\mathcal{W}|\}$. Recall that its decoder is defined as

$$w_t = \mathcal{S}^{\mathrm{inv}}(\boldsymbol{P}_t, \zeta_t) := \pi_t^{-1}\big(F^{-1}(U_t; \pi_t)\big),$$

where $\zeta_t = (\pi_t, U_t)$ with $U_t \sim U(0,1)$ and π_t being sampled uniformly at random from all permutations on $\mathcal{W}$. Following the strategy in [34, 47], (π_t, U_t)'s are jointly independent across the token sequence. For a given permutation π, $F(\cdot, \pi)$ above denotes the CDF under permutation: $F(x;\pi) = \sum_{w'\in\mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w')\leq x\}}$. By construction, the inverse transform watermark is unbiased for sampling from the NTP distribution $\boldsymbol{P}$. To see the unbiasedness directly, for any token w, note that $\mathcal{S}^{\mathrm{inv}}(\boldsymbol{P}, \zeta) = w$ if and only if

$$\sum_{w'\in\mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w')<\pi(w)\}} \leq U \leq \sum_{w'\in\mathcal{W}} P_{w'} \cdot \mathbf{1}_{\{\pi(w')\leq\pi(w)\}}, \tag{23}$$

which occurs with probability P_w since the interval above has length P_w.

4.1. *Main results.* To utilize the framework in Section 2 for the inverse transform watermark, we need to construct a pivotal statistic $Y_t = Y(U_t, \pi_t)$ such that its distribution is known under the null, and meanwhile should capture the dependence between U_t and π_t under the alternative for any NTP distribution. Under H_0, U_t is independent of the human-written token w_t, and $\pi_t(w_t)$ is uniformly distributed over the vocabulary $\mathcal{W}$. Under H_1, in contrast, a larger U_t value suggests a larger value of $\pi_t(w_t)$ in distribution for the watermarked text,

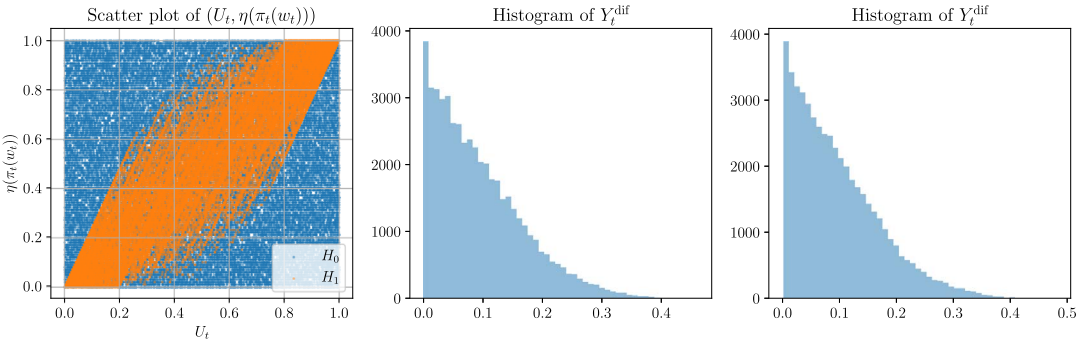


FIG. 5. *Left*: *The scatter plot of* 5×10^4 *i.i.d. observations from* $(U_t, \eta(\pi_t(w_t)))$ *when* $|\mathcal{W}| = 100$ *and the top five probabilities in* $\boldsymbol{P}_t$ *are fixed as* $(0.2, 0.2, 0.1, 0.05, 0.05)$ *and the rest probabilities are uniformly distributed. Middle*: *Histogram of* 10^4 *i.i.d. observations of* Y_t^{dif} *when the underlying NTP distribution is* $\boldsymbol{P}_t$. *Right*: *The same setting as the middle panel except that the probabilities except for the top five of* $\boldsymbol{P}_t$ *follow normalized* Pow(0.5, 1) *values*.

Non_parametric_Lecture_Note_6.pdf

which can be gleaned from (23). See the left panel of Figure 5 for a simulation that illustrates the distinct behaviors of $(U_t, \eta(\pi_t(w_t))$ under H_0 and H_1.

This motivates the use of the following statistic for watermark detection [34, 47]:

$$Y_t^{\text{dif}} = |U_t - \eta(\pi_t(w_t))|, \quad \eta(i) := \frac{i-1}{|\mathcal{W}|-1},$$

where η normalizes an integer token to $[0, 1]$ for direct comparison with $U \in [0, 1]$. As is clear, this statistic is a pivot for our framework.

Technical challenge. To continue under our framework, we are faced with a difficulty in characterizing the distribution of the Y_t^{dif} under the alternative. Although it is clear from Figure 5 that Y_t^{dif} tends to be small under the alternative, the complex dependence of Y_t^{dif}'s distribution on the unknown NTP $\boldsymbol{P}_t$ under H_1 renders the evaluation of class-dependent efficiency in the framework generally intractable.

Formally, this technical challenge can be elucidated by Lemma 4.1 and its following remark.

LEMMA 4.1. *Under* H_0, *the CDF of* Y_t^{dif} *is*

$$F_0^{\text{dif}}(r) := \mathbb{P}_{H_0}(Y_t^{\text{dif}} \le r) = \frac{1}{|\mathcal{W}|} \sum_{i=1}^{|\mathcal{W}|} [\min\{\eta(i) + r, 1\} - \max\{\eta(i) - r, 0\}].$$

Under H_1, *the CDF of* Y_t^{dif} *is*

$$F_{1,\boldsymbol{P}_t}^{\text{dif}}(r) := \mathbb{P}_{H_1}(Y_t^{\text{dif}} \le r | \boldsymbol{P}_t) = \frac{1}{|\mathcal{W}|!} \sum_{\pi \in \Pi} \sum_{i=1}^{|\mathcal{W}|} |(a_{\pi,i-1}, a_{\pi,i}] \cap B(\eta(i), r)|,$$

where Π *collects all permutations on* $\mathcal{W}$, $a_{\pi,i} = \sum_{j=1}^{i} P_{t,\pi(j)}$, $B(v, r) = \{x \in [0, 1] : |x - v| \le r\}$ *and* $|\cdot|$ *represents the length of an interval*.

In Lemma 4.1, the first part in particular shows that Y_t^{dif} is a pivotal statistic under H_0. The second part provides the explicit CDF of Y_t^{dif} under H_1 conditional on $\boldsymbol{P}_t$.

Nonetheless, Lemma 4.1 also shows that the distribution formula for Y_t^{dif} is by no means simple under the alternative hypothesis H_1: the dependence on $\boldsymbol{P}_t$ is intricate. The intricacy arises due to the nature of $\mathcal{S}$, whose definition involves all permutations Π of $\mathcal{W}$, which also

complicates the distribution formula for $Y_t^{\mathrm{dif}} = |U_t - \eta(\pi_t(w_t))|$ where $w_t = \mathcal{S}(P_t, (U_t, \pi_t))$. Such complexity in the distribution formula poses significant technical challenges in evaluating the effectiveness of any test procedure under the alternative hypotheses H_1.

Considering these intricacies, our goal of deriving an optimal score function h based on Y_t^{dif} seems challenging at first glance.

Asymptotic distributions. Fortunately, we discover a surprising result that simplifies the understanding of the distribution of Y_t^{dif} under the alternative H_1, which facilitates later derivation of the optimal score function h. Roughly speaking, the conditional distribution of Y_t^{dif} given $\boldsymbol{P}_t$ under H_1 primarily depends on the largest values among $\boldsymbol{P}_t$'s coordinates, especially when the vocabulary size $|\mathcal{W}|$ is large. For illustration, histograms in Figure 5 compare Y_t^{dif} under H_1 for different $\boldsymbol{P}_t$ and $\boldsymbol{P}_t'$ scenarios where $|\mathcal{W}| = 100$. Despite differences in $\boldsymbol{P}_t$ and $\boldsymbol{P}_t'$, each sharing the same top five coordinates but differing elsewhere, the conditional distribution of Y_t^{dif} remains remarkably similar.

We now mathematically formalize this empirical observation under a special case where there is one token that takes predominantly high probability in $\boldsymbol{P}_t$. This scenario, being the most basic case in theory, offers practical insights as empirically, many LLMs often concentrate most of $\boldsymbol{P}_t$'s probability mass on a single token [21]. Through our analysis, we derive the limiting distribution of Y_t^{dif} given $\boldsymbol{P}_t$ when the vocabulary size $|\mathcal{W}| \to \infty$. Formally, this means that we are effectively examining a sequence of $\boldsymbol{P}_t$, which is indexed by $|\mathcal{W}|$, and investigate the limiting conditional distribution of Y_t^{dif} given $\boldsymbol{P}_t$ as we move along that sequence. Yet when the context makes it clear, we suppress the dependence of $\boldsymbol{P}_t$ on the index $|\mathcal{W}|$ for notation simplicity.

In below, we always use $P_{t,(i)}$ to refer to the ith largest coordinates of $\boldsymbol{P}_t$ for every vector $\boldsymbol{P}_t$ and integer i between 1 and $|\mathcal{W}|$.

THEOREM 4.1. *Under* H_0,

$$\lim_{|\mathcal{W}|\to\infty} \mathbb{P}_{H_0}\big(Y_t^{\mathrm{dif}} \le r\big) = 1 - (1-r)^2 \quad \textit{for any } r \in [0,1].$$

Under H_1, *assuming that* $\lim_{|\mathcal{W}|\to\infty} P_{t,(1)} = 1 - \Delta$ *and* $\lim_{|\mathcal{W}|\to\infty} \log|\mathcal{W}| \cdot P_{t,(2)} = 0$ *hold, then*

$$\lim_{|\mathcal{W}|\to\infty} \mathbb{P}_{H_1}\big(Y_t^{\mathrm{dif}} \le r \,|\, \boldsymbol{P}_t\big) = 1 - \left(1 - \frac{r}{1-\Delta}\right)^2 \quad \textit{for any } r \in [0, 1-\Delta].$$

Theorem 4.1 formalizes this observed phenomenon, providing a simplified distribution characterization of the statistics Y_t^{dif} under H_1, assuming that the token probability $\boldsymbol{P}_t$ is highly concentrated at a single token, under the large vocabulary size limit $|\mathcal{W}| \to \infty$. The limiting distribution is determined by a single scalar $1 - \Delta$ that corresponds to the top token probability.

Notably, in this limit $|\mathcal{W}| \to \infty$, the statistics Y_t^{dif} has a different support under H_0 and under H_1. By exploiting this distinction, we proceed to identify the optimal score h under our framework, which achieves infinite power in the limit $|\mathcal{W}| \to \infty$ in distinguishing the hypotheses H_0 and H_1.

Optimal score function. Following our previous discussions, we focus on scenarios where an LLM's probability distribution, $\boldsymbol{P}_t$, is primarily concentrated on a single token.

We model $\boldsymbol{P}_t$ using a belief class $\overline{\mathcal{P}}_\Delta$, a subset of the Δ-regular class $\mathcal{P}_\Delta$. Unlike $\mathcal{P}_\Delta$, which only restricts the highest token probability to $1 - \Delta$, the belief class $\overline{\mathcal{P}}_\Delta$ also sets a

limit on the second highest token probability, using a threshold $\varepsilon_{|\mathcal{W}|}$:[16]

$$\overline{\mathcal{P}}_\Delta = \{\boldsymbol{P} : P_{(1)} \le 1 - \Delta,\, P_{(2)} \le \varepsilon_{|\mathcal{W}|}\}. \tag{24}$$

In our subsequent asymptotic analysis $|\mathcal{W}| \to \infty$, we assume that $\log|\mathcal{W}| \cdot \varepsilon_{|\mathcal{W}|} \to 0$, from which we can utilize Theorem 4.1 to simplify our distributional characterization of Y_t^{dif}.

This leads to the definition of a limit efficiency measure, $\overline{R}_\Delta$, whose definition is based on our efficiency measure $R_{\overline{\mathcal{P}}_\Delta}$ under our framework in Section 2:

$$\overline{R}_\Delta(h) = \liminf R_{\overline{\mathcal{P}}_\Delta}(h), \tag{25}$$

where lim inf denotes the large vocabulary size limit where $|\mathcal{W}| \to \infty$ and $\log|\mathcal{W}| \cdot \varepsilon_{|\mathcal{W}|} \to 0$. In plain words, $\overline{R}_\Delta(h)$ quantifies the efficiency of any score function h in the limit of large vocabulary size $|\mathcal{W}| \to \infty$, assuming that the distribution of the underlying LLM predominantly focuses on a single token.

We are ready to describe the optimal score function h that maximizes this efficiency measure $\overline{R}_\Delta(h)$. This optimal procedure is formally stated in Theorem 4.2.

THEOREM 4.2. *Fix* $\Delta \in (0, 1]$. *Let* $h^\star_{\mathrm{dif},\Delta} : [0, 1] \to \mathbb{R}$ *denote*

$$h^\star_{\mathrm{dif},\Delta}(r) = \log \frac{f_{\mathrm{dif},\Delta}(r)}{f_{\mathrm{dif},0}(r)} \quad \textit{where } f_{\mathrm{dif},\Delta}(r) = \frac{2}{1-\Delta} \cdot \max\left\{1 - \frac{r}{1-\Delta}, 0\right\}.$$

Then $\lim_{M\to\infty} \overline{R}_\Delta([h^\star_{\mathrm{dif},\Delta}]_{[-M,M]}) = \infty$.

Theorem 4.2 identifies the score function $h^\star_{\mathrm{dif},\Delta}$ that achieves infinite power in distinguishing H_1 from H_0 in the large vocabulary limit $|\mathcal{W}| \to \infty$, assuming that the NTP distribution $\boldsymbol{P}_t$ focuses on a single token, formally described by $\overline{\mathcal{P}}_\Delta$ with $\log|\mathcal{W}| \cdot \varepsilon_{|\mathcal{W}|} \to 0$. The right panel of Figure 3 shows $h^\star_{\mathrm{dif},\Delta}$ for different values of Δ.

Our derivation of $h^\star_{\mathrm{dif},\Delta}$ is based on calculating a log-likelihood ratio between the hardest alternative within $\overline{\mathcal{P}}_\Delta$ and the null, using the asymptotic distribution formula described in Theorem 4.1. The log-likelihood ratio $h^\star_{\mathrm{dif},\Delta}$ achieves infinite power due to Y^{dif} having distinct supports under H_0 and H_1 in the limit, as demonstrated in Theorem 4.1.

Finally, the truncation in $[h^\star_{\mathrm{dif},\Delta}]_{[-M,M]}$ is mainly for technical reasons, stemming from a proof artifact.

4.2. *Proof of Theorem* 4.2. Fix $\Delta \in (0, 1]$. To prove Theorem 4.2, our goal is to establish the limit: $\lim_{M\to\infty} \overline{R}_\Delta([h^\star_{\mathrm{dif},\Delta}]_{[-M,M]}) = \infty$. Our strategy involves establishing an effective lower bound for $\overline{R}_\Delta(h)$, as detailed in Lemma 4.2. This involves introducing a function $F_{\mathrm{dif},\Delta}$ for each $\Delta \in [0, 1]$, defined as $F_{\mathrm{dif},\Delta}(r) = 1 - \max\{1 - \frac{r}{1-\Delta}, 0\}^2$ for any $r \in [0, 1]$. By Theorem 4.1, this corresponds to the limiting CDF of Y^{dif} given $\boldsymbol{P}_t$, assuming that $\boldsymbol{P}_t \in \overline{\mathcal{P}}_\Delta$.

LEMMA 4.2. *For any function* h *that is nondecreasing, Lipschitz-continuous on* $[0, 1]$, *there is the lower bound*:

$$\overline{R}_\Delta(h) \ge -\left[\int_0^1 h(r) F_{\mathrm{dif},0}(\mathrm{d}r) + \log \int_0^1 \mathrm{e}^{-h(r)} F_{\mathrm{dif},\Delta}(\mathrm{d}r)\right]. \tag{26}$$

[16]For simplicity, we omit the notation indicating the dependence of $\overline{\mathcal{P}}_\Delta$ on $\varepsilon_{|\mathcal{W}|}$ when the context makes this clear.

The proof of Lemma 4.2 is deferred to Section 4.2.1.

Now we complete the proof of Theorem 4.2. Let $G(h)$ denote the RHS of inequality (26). Also, let $\mu_{\mathrm{dif},\Delta}$ denote the probability measure associated with the CDF $F_{\mathrm{dif},\Delta}$. By Donsker–Varadhan representation [19], we deduce

$$\sup_h G(h) = \sup_h \left[-\int_0^1 h(r) F_{\mathrm{dif},0}(\mathrm{d}r) - \log \int_0^1 \mathrm{e}^{-h(r)} F_{\mathrm{dif},\Delta}(\mathrm{d}r) \right] = D_{\mathrm{KL}}(\mu_{\mathrm{dif},0}, \mu_{\mathrm{dif},\Delta}) = \infty,$$

where the last identity holds because the two probability measures $\mu_{\mathrm{dif},0}$ and $\mu_{\mathrm{dif},\Delta}$ have different support, with $\mu_{\mathrm{dif},0}$ supported on $[0, 1]$ and $\mu_{\mathrm{dif},\Delta}$ on $[0, 1-\Delta]$ for $\Delta > 0$. This supremum $\sup_h G(h)$ is obtained for h as the log-likelihood ratio:

$$h^\star_{\mathrm{dif},\Delta} = \log \frac{\mathrm{d}\mu_{\mathrm{dif},\Delta}}{\mathrm{d}\mu_{\mathrm{dif},0}} = \log \frac{f_{\mathrm{dif},\Delta}}{f_{\mathrm{dif},0}}.$$

Despite $G(h^\star_{\mathrm{dif},\Delta}) = \infty$, we are not able to use Lemma 4.2 to conclude that $\overline{R}_\Delta(h^\star_{\mathrm{dif},\Delta}) = \infty$. This is because the function $h^\star_{\mathrm{dif},\Delta}$, despite being nonincreasing, is neither uniformly bounded nor Lipschitz-continuous on $\mathbb{R}$, thus not meeting the requirements of Lemma 4.2.

Nonetheless, after truncating $h^\star_{\mathrm{dif},\Delta}$ to $h_M = [h^\star_{\mathrm{dif},\Delta}]_{[-M,M]}$, the function h_M is nondecreasing, Lipschitz-continuous, and uniformly bounded on $\mathbb{R}$ for any $M < \infty$. This allows us to use Lemma 4.2 to conclude that for every $M < \infty$: $\overline{R}_\Delta(h_M) \geq G(h_M)$. Applying the limit $M \to \infty$ on both sides, and leveraging Lebesgue's dominated convergence theorem along with Fatou's lemma, we deduce

$$\lim_{M\to\infty} \overline{R}_\Delta\big([h^\star_{\mathrm{dif},\Delta}]_{[-M,M]}\big) = \lim_{M\to\infty} \overline{R}_\Delta(h_M) \geq \lim_{M\to\infty} G(h_M) \geq G\big(h^\star_{\mathrm{dif},\Delta}\big) = \infty.$$

This yields Theorem 4.2 as desired.

4.2.1. *Proof of Lemma 4.2.* We are interested in lower bounding $\overline{R}_\Delta(h) = \liminf R_{\overline{\mathcal{P}}_\Delta}(h)$. Recall that its definition is given by

$$\overline{R}_\Delta(h) = -\limsup \inf_{\theta\geq 0} \sup_{\boldsymbol{P}\in\overline{\mathcal{P}}_\Delta} \big[\theta \mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-\theta h(Y^{\mathrm{dif}})}\big]\big]. \tag{27}$$

Our lower bound strategy relies on first connecting $\overline{R}_\Delta(h)$ to an intermediate quantity that

$$\overline{L}(h, \Delta') := \limsup \sup_{\boldsymbol{P}\in\mathcal{Q}_{\Delta'}} \big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big],$$

where $\mathcal{Q}_{\Delta'} := \{\boldsymbol{P} : P_{(1)} = 1 - \Delta', \log|\mathcal{W}| \cdot P_{(2)} \leq \varepsilon_{|\mathcal{W}|}\}$. This connection is given in Lemma 4.3.

LEMMA 4.3. *For every function h Lipschitz-continuous on $[0, 1]$, we have*

$$\overline{R}_\Delta(h) \geq -\sup_{\Delta'\geq\Delta} \overline{L}(h, \Delta').$$

Lemma 4.3 implies that maximizing $\overline{R}_\Delta(h)$ is related to the minimax optimization: $\inf_h \sup_{\Delta'\geq\Delta} \overline{L}(h, \Delta')$. This minimax formulation is closely related to our original minimax formulation (14), yet it replaces the $|\mathcal{W}|$-dimensional variable $\boldsymbol{P}$ with a scalar Δ' after taking asymptotics (Theorem 4.1).

PROOF OF LEMMA 4.3. By definition, we have

$$\inf_{\theta\geq 0} \sup_{\boldsymbol{P}\in\overline{\mathcal{P}}_\Delta} \big[\theta \mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-\theta h(Y^{\mathrm{dif}})}\big]\big] \leq \sup_{\boldsymbol{P}\in\overline{\mathcal{P}}_\Delta} \big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big].$$

After we substitute this inequality into (27), we obtain a lower bound

$$\begin{aligned}\overline{R}_{\Delta}(h) &\geq -\limsup \sup_{\boldsymbol{P}\in\overline{\mathcal{P}}_{\Delta}} \big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big] \\ &= -\limsup \sup_{\Delta'\geq\Delta} \sup_{\boldsymbol{P}\in\mathcal{Q}_{\Delta'}} \big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big],\end{aligned} \tag{28}$$

where the second identity holds because $\overline{\mathcal{P}}_{\Delta} = \bigcup_{\Delta'\geq\Delta} \mathcal{Q}_{\Delta'}$ by definition.

We wish to swap the order of $\sup_{\Delta'\geq\Delta}$ and lim sup in (28) to reach the conclusion of Lemma 4.3. Lemma 4.4 justifies such a swap as valid.

LEMMA 4.4. *For any Lipschitz-continuous function h on $[0, 1]$, we have*

$$\begin{aligned}&\limsup \sup_{\Delta'\geq\Delta} \sup_{\boldsymbol{P}\in\mathcal{Q}_{\Delta'}} \big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big] \\ &= \sup_{\Delta'\geq\Delta} \limsup \sup_{\boldsymbol{P}\in\mathcal{Q}_{\Delta'}} \big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big].\end{aligned}$$

By swapping $\sup_{\Delta'\geq\Delta}$ and lim sup in (28), we obtain

$$\overline{R}_{\Delta}(h) \geq -\sup_{\Delta'\geq\Delta} \limsup \sup_{\boldsymbol{P}\in\mathcal{Q}_{\Delta'}} \big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big] = -\sup_{\Delta'\geq\Delta} \overline{L}(h, \Delta').$$

This completes the proof of Lemma 4.3. □

Lemma 4.3 reduces the problem of finding a lower bound on $\overline{R}_{\Delta}(h)$ to evaluating $\sup_{\Delta'\geq\Delta} \overline{L}(h, \Delta')$. To do so, we first derive an explicit expression for $\overline{L}(h, \Delta')$.

LEMMA 4.5. *For any Lipschitz-continuous h on $[0, 1]$, we have*

$$\overline{L}(h, \Delta') = \int_0^1 h(r) F_{\mathrm{dif},0}(\mathrm{d}r) + \log \int_0^1 \mathrm{e}^{-h(r)} F_{\mathrm{dif},\Delta'}(\mathrm{d}r).$$

The proof of Lemma 4.5 is deferred to Supplement C.4. Here, we give some intuition behind its proof. Theorem 4.1 states that as $|\mathcal{W}| \to \infty$, for any sequence of $\boldsymbol{P} \in \mathcal{Q}_{\Delta'}$ indexed by $\mathcal{W}$, the CDF of Y^{dif} under H_0 converges to $F_{\mathrm{dif},0}$, while its CDF under H_1 given $\boldsymbol{P}$ converges to $F_{\mathrm{dif},\Delta'}$. This results in a convergence statement that holds for any sequence of $\boldsymbol{P} \in \mathcal{Q}_{\Delta'}$, and for any continuous function h uniformly bounded on $\mathbb{R}$:

$$\lim\big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big] = \int_0^1 h(r) F_{\mathrm{dif},0}(\mathrm{d}r) + \log \int_0^1 \mathrm{e}^{-h(r)} F_{\mathrm{dif},\Delta'}(\mathrm{d}r).$$

Lemma 4.5 strengthens this by ensuring such convergence can be made uniform across the sequence:

$$\begin{aligned}\overline{L}(h, \Delta') &= \limsup \sup_{\boldsymbol{P}\in\mathcal{Q}_{\Delta'}} \big[\mathbb{E}_0 h\big(Y^{\mathrm{dif}}\big) + \log \mathbb{E}_{1,\boldsymbol{P}}\big[\mathrm{e}^{-h(Y^{\mathrm{dif}})}\big]\big] \\ &= \int_0^1 h(r) F_{\mathrm{dif},0}(\mathrm{d}r) + \log \int_0^1 \mathrm{e}^{-h(r)} F_{\mathrm{dif},\Delta'}(\mathrm{d}r).\end{aligned}$$

The caveat is that the above uniform limit applies only to functions h that are Lipschitz-continuous and uniformly bounded on $\mathbb{R}$, as required in Lemma 4.5.

Finally, with Lemma 4.5 we are able to derive an explicit expression of the supremum $\sup_{\Delta'\geq\Delta} \overline{L}(h, \Delta')$.

LEMMA 4.6. *For any nonincreasing and Lipschitz-continuous h on* $[0, 1]$,

$$\sup_{\Delta' \geq \Delta} \overline{L}(h, \Delta') = \overline{L}(h, \Delta) = \int_0^1 h(r) F_{\mathrm{dif},0}(\mathrm{d}r) + \log \int_0^1 \mathrm{e}^{-h(r)} F_{\mathrm{dif},\Delta}(\mathrm{d}r).$$

PROOF OF LEMMA 4.6. Using integration by parts, we first obtain the equation

$$\int_0^1 \mathrm{e}^{-h(r)} F_{\mathrm{dif},\Delta'}(\mathrm{d}r) = \mathrm{e}^{-h(1)} - \int_0^1 F_{\mathrm{dif},\Delta'}(r) \mathrm{e}^{-h(r)}(-h)(\mathrm{d}r).$$

As a consequence, the above integral's value is nonincreasing as Δ' increases because $F_{\mathrm{dif},\Delta'}(r)$ is nondecreasing in Δ' for any r within the interval $[0, 1]$.

By Lemma 4.5, this further implies that $\overline{L}(h, \Delta')$ is nondecreasing in Δ'. Thus, $\sup_{\Delta' \geq \Delta} \overline{L}(h, \Delta') = \overline{L}(h, \Delta)$. The conclusion of Lemma 4.6 then follows. □

We now complete the proof of Lemma 4.2. By Lemma 4.3 and Lemma 4.6, we obtain

$$\overline{R}_\Delta(h) \geq -\sup_{\Delta' \geq \Delta} \overline{L}(h, \Delta') = -\int_0^1 h(r) F_{\mathrm{dif},0}(\mathrm{d}r) - \log \int_0^1 \mathrm{e}^{-h(r)} F_{\mathrm{dif},\Delta}(\mathrm{d}r).$$

This yields the lower bound as desired.

5. Experiments. This section highlights our framework's effectiveness through synthetic and real-data experiments, showcasing the practical utility of our proposed methods for detecting watermarks.

5.1. *Synthetic studies*. In our simulation, we generate a vocabulary $\mathcal{W}$ of size 1000 and assess Type I and II errors in watermark detection methods for generated token sequences. These methods include $h_{\mathrm{ars}}(Y^{\mathrm{gum}})$, $h_{\log}(Y^{\mathrm{gum}})$, $h_{\mathrm{ind},\mathrm{e}^{-1}}(Y^{\mathrm{gum}})$, $h^{\star}_{\mathrm{gum},\Delta}(Y^{\mathrm{gum}})$ for Gumbel-max watermark and $h_{\mathrm{neg}}(Y^{\mathrm{dif}})$, $h^{\star}_{\mathrm{dif},\Delta}(Y^{\mathrm{dif}})$ for the inverse transform watermark.

Our initial investigation is on Type I error control for finite token sequences, recognizing that the guarantees for Type I error control follow from the central limit theorem and hold asymptotically when the sequence length $T \to \infty$. For a given text length T, we generate 5000 samples of *unwatermarked* word tokens sequences. We uniformly sample each unwatermarked token from the vocabulary $\mathcal{W}$ in our experiments. Although different sampling strategies could be employed, due to the pivotal property of our test statistics, we anticipate that they would lead to similar Type I error rates.

Throughout 5000 repeated experiments, we compute the average Type I error, with the findings presented in the leftmost column of Figure 6. These results reveal that empirically, the Type I errors generally align with the nominal level of 0.05, fluctuating between 0.04 and 0.06. This performance confirms our theoretical expectations regarding Type I error control across most scenarios. The only exception is the score function $h_{\mathrm{ind},\delta}(Y^{\mathrm{gum}})$, achieving close to the nominal 0.05 level only for token sequences over 300 in length, where asymptotic effects emerge.

Next, we examine the effectiveness of these methods in controlling Type II errors on *watermarked* sequences. Performing this examination involves specifying how each $\boldsymbol{P}_t$ is generated under H_1. In our simulation, we model LLM's NTP $(\boldsymbol{P}_t)$ distributions as spike distributions: we set its largest probability as $P_{t,(1)} = 1 - \Delta_t$, where $\Delta_t > 0$ is i.i.d. sampled from $[a, b]$ with $a = 10^{-3}$,[17] and $b = 0.5$, and uniformly distribute the remaining probabilities so that $P_{t,(k)} = \Delta_t/(|\mathcal{W}| - 1)$ for $k > 1$. The choice of this setup is based on our theoretical

[17]This small number 10^{-3} is used to avoid H_1 merging with H_0.

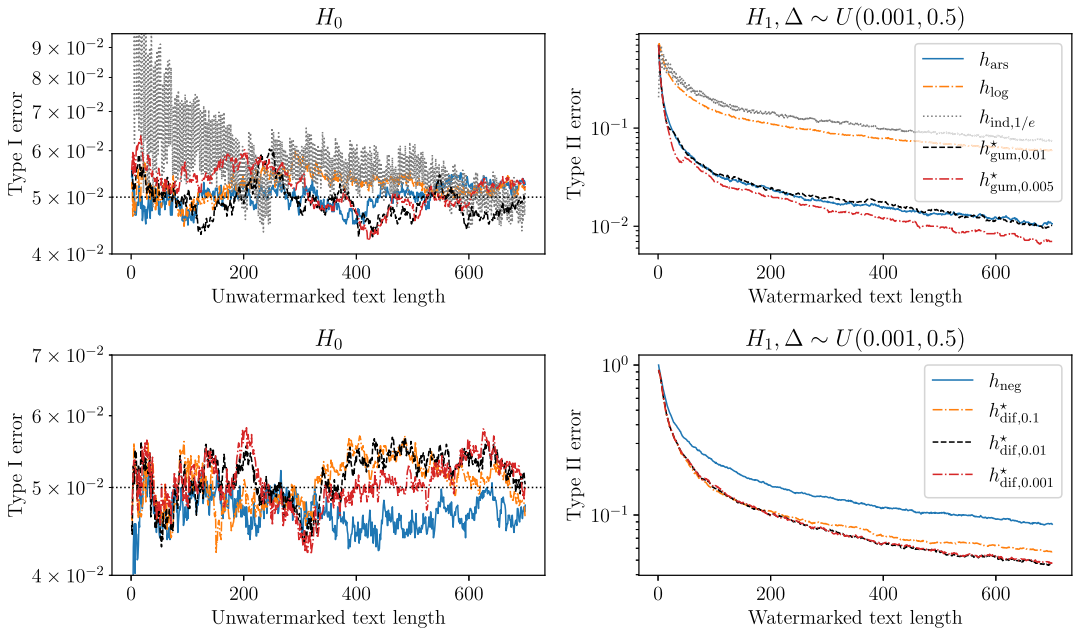


FIG. 6. *Average Type I and II errors* (*on a log scale*) *vs. text length on synthetic data sets for Gumbel-max* (*top*) *and inverse transform watermarks* (*bottom*).

development. We also tested alternative b values of 0.1, 0.3 and 0.7, but given the results are similar, we will include those details in the Supplementary Material.

Throughout 5000 repeated experiments, we compute the average Type II error and present the results in the rightmost column of Figure 6. Designed using the optimality criterion through our framework, both $h^\star_{\text{gum},\Delta}(Y^{\text{gum}})$ for Gumbel-max and $h^\star_{\text{dif},\Delta}(Y^{\text{dif}})$ for inverse transform outperform other watermark detection techniques in reducing Type II errors. Additionally, empirical performance rankings are consistent with our theoretical prediction in Theorem 3.3: $h_{\text{ars}}(Y^{\text{gum}})$ has lower Type II error, followed by $h_{\log}(Y^{\text{gum}})$, with $h_{\text{ind},\text{e}^{-1}}(Y^{\text{gum}})$ having the highest Type II error.

5.2. *Real-world examples*. In this section, we conduct an empirical analysis of watermark detection methods for text sequences from large token generation models such as OPT-1.3B [75] and Sheared-LLaMA-2.7B [69]. Specifically, we evaluate Type I errors on the *unwatermarked* texts directly from large token generation models and Type II errors with texts from the same models, but with watermarks incorporated. This experimental setup essentially follows the approach described in the existing literature [31] and is further detailed in Supplement D.2 for completeness.

We present our numerical results with the OPT-1.3B models in Figure 7 (similar results for Sheared-LLaMA-2.7B are documented in Supplement D.2). Notably, both the average Type I error (leftmost column of Figure 7) and Type II error (rightmost column of Figure 7) curves are computed based on 500 repeated experiments. Though the setup (mainly the way each $\boldsymbol{P}_t$ is generated) is slightly different from our simulation studies, the conclusions are quite similar.

First, in most scenarios, all the detection methods maintain Type I errors within 0.04 to 0.06, aligning well with the expected 0.05 level, particularly in texts over 100 tokens long, as observed in our numerical experiments. Second, our methods, $h^\star_{\text{gum},\Delta}(Y^{\text{gum}})$ for Gumbel-max and $h^\star_{\text{dif},\Delta}(Y^{\text{dif}})$ for inverse transform, excel in minimizing Type II errors over other score functions. For Gumbel-max watermarks, the performance rankings follow our

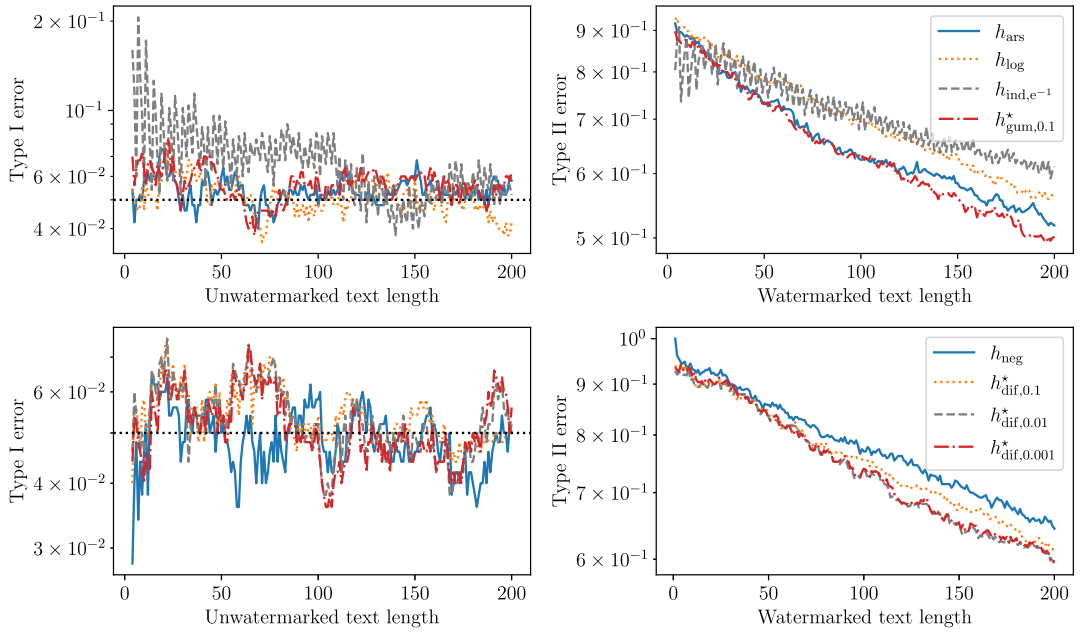


FIG. 7. *Average Type I and II errors* (*on a log scale*) *vs. text length on* C4 *data sets* [50] *and the* OPT-1.3B *model for Gumbel-max* (*top*) *and inverse transform watermarks* (*bottom*).

Theorem 3.3's predictions: $h_{\text{ars}}(Y^{\text{gum}})$ records a lower Type II error, then $h_{\log}(Y^{\text{gum}})$, with $h_{\text{ind},e^{-1}}(Y^{\text{gum}})$ the highest type II error.

6. Discussion. In this paper, we have introduced a statistical framework for reasoning about watermarks for LLMs from a hypothesis testing viewpoint. Our framework offers techniques and concepts for analyzing the asymptotic test efficiency of watermark detection by evaluating Type I and Type II errors of the problem of watermark detection. Specifically, to provide a formal approach to comparing different detection rules, we have introduced class-dependent efficiency, which further reduces the comparison problem to a minimax optimization problem. We have applied this framework to two representative unbiased watermarks, namely, the Gumbel-max watermark and the inverse transform watermark. Our results include the derivation of optimal detection rules, which are shown to have competitive or superior performance compared to existing detection rules.

Our work opens up several research directions in this emerging line of statistical research on watermarks for LLMs. Perhaps the most immediate question is how to choose Δ for the distribution class $\mathcal{P}_\Delta$ when the knowledge of the underlying token distributions is limited. For instance, it would be valuable to explore methods for enhancing detection power when Δ is known in a distributional sense (see the discussion in Supplement D.4). Another approach is to extend Proposition 2.1 to multiple distribution classes in recognition of the heterogeneity of NTP distributions across the token sequence. This can be done, for example, by adaptively clustering the empirical distributions of $\boldsymbol{P}_t$ into several clusters and using these to represent the distribution classes. Moreover, it would be a welcome advance to offer guidelines on how to choose pivotal statistics, which perhaps would allow the framework to maximize its utility. In this paper, we fix the pivotal statistics from the beginning. However, we show that it is fundamentally difficult to identify optimal pivotal statistics that maximize our efficiency notion. The challenge lies in the *nonconvex* nature of the set of pivotal statistics, making the selection problem a *nonconvex* optimization problem that is currently beyond our reach (see Supplement A.3). Finally, given that an unbiased watermark is probabilistically similar to

sampling from a multinomial distribution, it would be of interest to consider other sampling schemes such as the alias method [64, 65] to develop new categories of watermarks.

More broadly, we wish to remark on several possible extensions of our framework for improving watermark detection and design. Our framework would be enhanced if different watermarks with their own detection rules could be compared on a fair and comprehensive basis. Different watermarks might work best in different regimes and, therefore, directly comparing their class-dependent efficiency rates would be very sensitive to the choice of the distribution class. Another direction to extend our framework is to take distribution classes that are adaptive to NTP distributions in the real-world deployment of LLMs. This adaptivity has the potential to make the minimax formulation (14) better capture the empirical performance. A possible approach is to model the empirical distribution of the NTP distributions $\boldsymbol{P}_t$ using Zipf's law [79] or other laws. A challenge here, however, might arise from the fact that the distribution class is no longer permutation invariant. Finally, an interesting extension is to have varying score functions across the token sequence, as opposed to having the same h in the test statistic (10). This flexibility might enhance the power of watermark detection by taking into account the heterogeneity of the NTP distributions from token to token in generation.

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SUPPLEMENTARY MATERIAL

Supplement to "A statistical framework of watermarks for large language models: Pivot, detection efficiency and optimal rules" (DOI: 10.1214/24-AOS2468SUPPA; .pdf). The detailed proofs of theoretical results and experimental specifics are provided in the Supplementary Material [36].

Source codes to "A statistical framework of watermarks for large language models: Pivot, detection efficiency and optimal rules" (DOI: 10.1214/24-AOS2468SUPPB; .zip). The source codes are provided in the Supplementary Material [37].

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